SAFEWATER LINES / YOUR VOICE

Overview

Understanding the survey, the notebook and the company implications

A complete illustrated reading guide

327
response records

30
rating questions

1,892
written answers

The central interpretation

Protect strong pride, contribution and support within teams. Test more dependable recognition, people decisions and cooperation across teams.

Includes every notebook cell, all 30 survey items, 13 figures, readable reference tables and an evidence-linked action pathway.

Prepared 29 October 2026

Confidential employee-survey analysis. Aggregate results only. Proposed actions are for organisational review.

Contents and reading routes

Start with the visual overview for the main findings. Read Sections 1-5 for the data and code foundations, Sections 6-15 for the analyses, and Sections 16-18 for action and limitations. The appendix explains all 146 cells.

Contents and reading routes 2
Visual overview: what the company can use 3
How to use this illustrated guide 4
1. What data are being analysed? 4
2. What the question codes mean 5
3. How the Python code works 6
4. Cleaning and derived variables 8
5. How to interpret the main statistics 9
6. Who responded, and why that affects interpretation 10
7. What all 30 questions show 11
8. Reliability, correlations and what they can tell us 18
9. Group comparisons and response-style sensitivity 20
10. The single-choice HR priority 23
11. How written answers become themes 24
12. The two principal organisational questions 28
13. Additional insights from checking the same records 31
14. What the topic network adds 32
15. Why the notebook does not need every advanced method 34
16. What the company can do with the findings 34
17. How to know whether the company benefited 37
18. Reproducibility audit and current limitations 38
19. Sources and related project outputs 38
Appendix: every notebook cell explained 39

Visual overview: what the company can use

Positive attachment, uneven everyday experience

Pride is 92.4% favourable and contribution clarity 95.7%. Recognition is 72.5%, consistent people decisions 74.6%, and cross-team cooperation 82.3%. These are internal comparisons, without an external pass/fail benchmark.

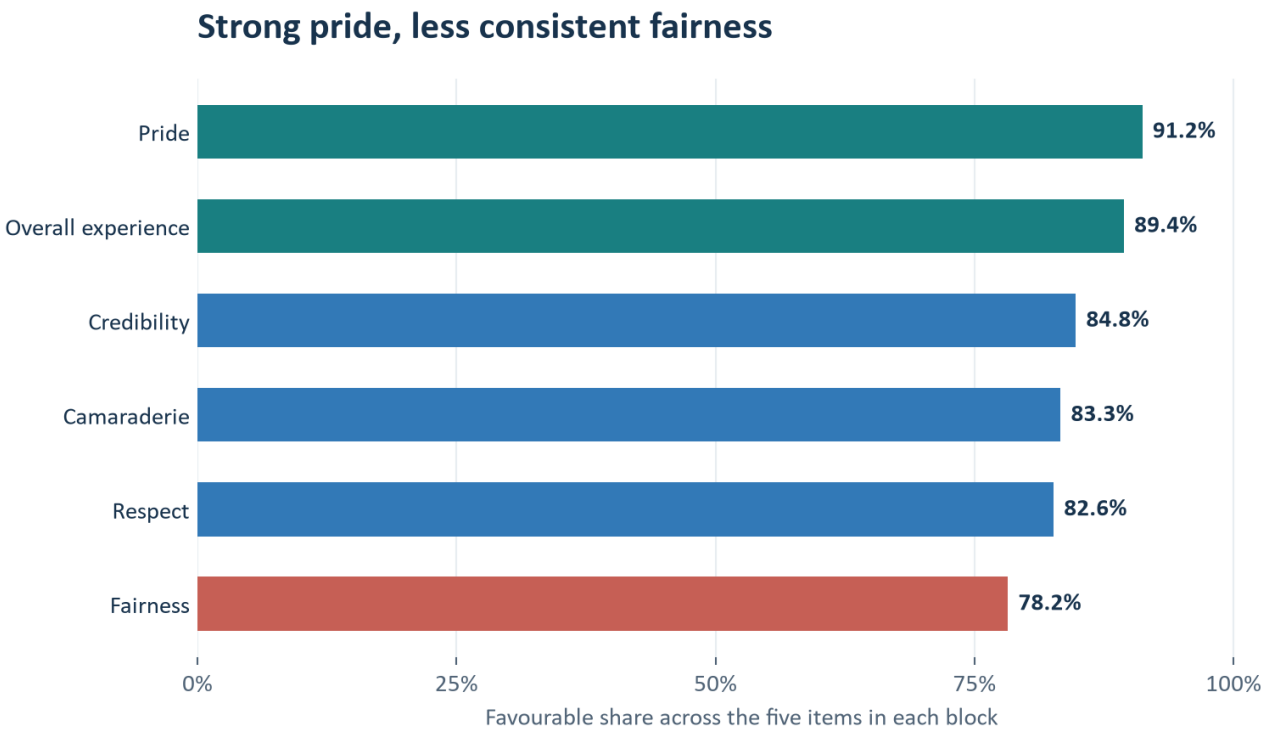


Figure 1. Block summaries describe item answers across the 327 responding records. Favourable means 4-5; the blocks are supplied questionnaire labels, not independently validated scales.

Three implications: preserve purpose and helpful relationships; improve consistency in specific daily practices; establish operational measures before promising a business benefit.

How certain is this? The ratings are complete and reproducible. Workforce response rate is unknown, comments are assistant-interpreted, and the survey does not establish causes or verify incidents.

Prepared 29 October 2026. This is a teaching companion to `notebooks/00_data_understanding_eda.ipynb`, with a source audit and additional descriptive checks. It explains the analytical choices, how to read the outputs, what they suggest for the company, and what remains unknown. The source workbook and notebook were not edited for this explanation.

The central finding is that respondents report a strong foundation of pride, contribution and support within their teams. They report less consistent recognition, fairness of people decisions and support across team boundaries. A practical response is to test clearer handovers, decision criteria, useful feedback and visible action on employee concerns, while preserving the existing strengths.

This is a working interpretation of the responding records. The survey does not establish causes, verified incidents, actual employee turnover or financial benefits from any proposed intervention.

How to use this illustrated guide

Sections 1-5 explain the source, code and measures. Sections 6-13 interpret findings and business implications. Sections 14-17 explain the network, method choices, actions and evaluation. The appendix describes each of the 146 cells.

The notebook has two layers. Its first 12 numbered sections prepare the survey and perform exploratory data analysis, or EDA. The later Steps 1-6 establish interpretation rules, code the comments, integrate the findings, explore the two selected priorities and draft the implementation pathway. Some early wording describes later work as still pending even though subsequent cells now implement it. Read the later steps for the current analytical status.

1. What data are being analysed?

The file `data/raw/Your Voice- Responses 21 OCT - FINAL.xlsx` is the source. The analysis uses its `SWL - MASTERDATA` sheet. The master has two header rows followed by **327 response records and 43 source columns**:

- Six demographic/identifying fields: location, designation, department, tenure, gender and an optional name.
- Thirty rating questions, arranged in six supplied blocks of five questions.
- Six written-answer questions.
- One question asking employees to select a single HR improvement priority.

After removing the dedicated name field and adding documented analysis variables, the processed survey has **327 rows and 56 columns**.

One row is a response record. We have not verified that every row corresponds to a different employee. There are no exact duplicate rows after excluding the name column, but absence of exact duplicates does not prove unique identity. The 9,810 rating answers are 327 records × 30 items. They are not 9,810 independent employees.

The eligible employee population was not supplied. Therefore **327 is a respondent count, not a response rate**. We cannot conclude that the responding records represent the whole workforce, nor calculate which locations or roles are overrepresented relative to their eligible populations.

Why the workbook inventory matters

The workbook also contains an older master, five language exports, and an HR pivot/mapping sheet. These are not interchangeable sources. The five language exports contain 330 response rows in total, while the selected master has 327. A row-level reconciliation explaining that difference is unavailable. The notebook correctly avoids inventing a story that three particular test responses were removed.

Excel converted some tenure bands in the language exports into dates or a date serial. A value intended as “5-10 years” must not be analysed as a calendar date. The selected master contains the corrected bands.

The HR pivot in `Sheet2` totals 328, which also differs from 327. The notebook borrows its option-to-theme mapping and recalculates the actual counts from the selected master. Reusing the pivot total would introduce an inconsistent denominator.

Quality and confidentiality

All 30 ratings and the five retained demographic fields are complete. Completeness does not prove that the form forced answers. There are **227 supplied names, or 69.4%** of the response records. The dedicated name column is removed from the processed CSV. Free-text comments and combinations of demographic details can still identify people, so removing that one column does not make the data anonymous.

The analysis keeps all 327 records. It reports patterns in aggregate, rather than using names to label employees or publishing identifying comments.

2. What the question codes mean

The codebook links each short identifier to the original wording. These are supplied questionnaire blocks, not independently established psychological scales.

Prefix	Supplied block	What its five questions concern
CRE	Credibility	Leadership honesty, keeping promises, direction, timely updates, listening and acting
RES	Respect	Manager support, resources/support/recognition, speaking up, appreciation, dignity
FAI	Fairness	Fair rewards, equal growth opportunities, consistent people decisions, workload fairness, understanding decisions
PRI	Pride	Company pride, contribution, intention to stay, business/customer impact, extra contribution
CAM	Camaraderie	Support within a team, collaboration across teams, belonging, access to other teams, celebrating success
OVR	Overall Employee Experience	Recommendation, growth, ability to perform, overall satisfaction, meaningful work

For example, FAI3 is the third fairness question: consistency of people-related decisions without favouritism or bias. FAI5 asks whether employees understand those decisions. These questions point to related but different management actions.

The six comment fields ask about leadership trust, feeling valued, fairness concerns, what to preserve, teamwork improvements and one desired change. The question context changes the meaning of an answer. “Nothing” in the fairness question can mean no concern; the same word in the preserve question can mean no valued aspect was identified.

3. How the Python code works

The notebook is an ordered sequence of explanatory cells and code cells. Markdown explains the reasoning. Code loads data, creates variables, calculates results and saves outputs. An .ipynb file also stores execution counts and displayed results. A displayed table or plot can survive from an earlier run, so the source and a fresh calculation are stronger evidence than a saved image alone.

The libraries

Tool	Its role here
pathlib.Path	Construct paths to input and output folders
pandas / pd	Load Excel, manage tabular data, group records and export CSVs
numpy / np	Numerical arrays, means, comparisons and matrix calculations
matplotlib.pyplot / plt	Create and save charts
seaborn / sns	Plot distributions, heatmaps, bars and boxplots
scipy.stats	Exploratory rank tests, multiplicity adjustment and contingency-table calculations
Counter, defaultdict, combinations	Count topics, retain contributor sets and enumerate topic pairs
hashlib.sha256	Detect changes in comment text or order before reusing the coding ledger
re, unicodedata	Normalize text and identify words/phrases
scipy.optimize.minimize	Arrange the topic network in two dimensions
IPython.display	Display tables, explanatory Markdown and links inside the notebook

ROOT finds the project folder. RAW points to the source workbook, OUT to processed data, TABLES to aggregate CSVs and FIGURES to exported charts. The path rule supports running from the repository root or its notebooks folder. It is not a general search for the project from any arbitrary working directory.

Expressions you see repeatedly

Expression	Plain meaning
`df` / `raw` / `txt`	Clean analysis table / loaded source table / stripped comment text
`df[ITEMS]`	Select the 30 rating columns
`.str.strip().str.lower()`	Remove edge whitespace and normalize capitalization
`.map(SCALE)`	Replace each recognized text label with its numeric score
`.eq(3)`, `.ge(4)`, `.le(2)`	Equal to 3, at least 4, at most 2
`.sum()` on Boolean values	Count the `True` values
`.mean()` on Boolean values	Calculate the fraction that are `True`
`axis=1` / `axis=0`	Work across columns within each row / down rows within each column
`.groupby(...)`	Collect records by a category, then summarize each group
`.agg(...)`	Calculate named summaries such as count, mean and median
`pd.crosstab(a, b)`	Count combinations of two categories
`.reindex(...)`	Align a result with a specified set/order of labels
`.explode('topics')`	Turn a multi-topic list into separate topic rows for counting
`.nunique()`	Count distinct values, such as source records rather than repeated mentions
`.loc[condition, columns]`	Select rows by a condition and choose their columns
`.where(...)` / `.fillna(...)`	Retain values conditionally / replace missing values
`.round(2)`	Round the displayed result; the main calculations retain precision
`assert ...`	Stop execution if a required consistency condition fails

For example, `df['RES4'].ge(4).sum()` counts favourable recognition answers. Dividing by the valid answer count and multiplying by 100 produces the favourable percentage.

4. Cleaning and derived variables

Rating labels

The source contains nine text variants of five response categories. Capitalization variants are normalized. The notebook maps `Usually` and the Hindi-form `Often` to 4, and `Almost always` and the Malayalam-form `Always` to 5. This is a documented harmonization choice. It does not independently establish that wording has exactly the same meaning in every language.

Numeric score	Harmonized response
1	Almost never
2	Rarely
3	Sometimes
4	Usually / Often
5	Almost always / Always

After mapping, all 9,810 rating cells contain a recognized value from 1 to 5. Score 3 means **Sometimes**, not "neutral" and not necessarily dissatisfaction.

Reporting groups

`region` groups cities into West, North, South, East and Overseas. `tenure` is an ordered category: 0-3, 3-5, 5-10, 10-15 and 15+ years. The band labels are supplied categories; the analysis does not turn them into exact years.

`level` approximates job level using designation keywords. Rule order matters: assistant manager must be recognized before a general manager keyword catches it. Labels such as "6 Sr Leadership" are category names, not a measured numerical distance from "3 Sr Executive." Ambiguous designations remain Unclear. HR should verify the hierarchy before using these groups for consequential comparisons.

`dept_group` retains departments with at least 10 respondents and combines smaller ones as Other. Other contains different functions and is not a single operating team. Combining records helps display but does not make them comparable in all respects.

`gave_name` records whether the optional name was supplied; `straightliner` records whether all 30 ratings are identical. These describe the response context. They do not establish truthfulness or bias.

Working scores

Each block mean, such as `FAI`, averages its five questions for each record. The overall reported block mean then averages those respondent means. Complete data and equal weighting make this equal to the average of all answers in that block.

`CULTURE_MEAN` averages the 25 CRE/RES/FAI/PRI/CAM items, excluding OVR. It is an analyst-created descriptive summary on a 1-5 scale. It is not an official Trust Index, a validated engagement measure or a percentage of

employees who are engaged. `PRI3`, intention to stay, is inside it, so correlating that composite with `PRI3` would contain a built-in part-whole relationship.

`fav_pct` is different: it is each record's percentage of favourable answers across **all 30** items. With complete data, its respondent average equals the pooled proportion of favourable item answers. Neither measure should be silently substituted for the other.

5. How to interpret the main statistics

Counts, percentages and denominators

For an item:

```
Favourable % = (number selecting 4 or 5 / valid answers to the item) × 100

Unfavourable % = (number selecting 1 or 2 / valid answers) × 100

Top-box % = (number selecting 5 / valid answers) × 100
```

Every item currently has 327 valid answers. Recognition RES4 has 237 favourable, 56 Sometimes and 34 unfavourable responses. Its favourable share is 237/327 = **72.5%**. The remaining 90 records include 56 Sometimes answers; calling all 90 “dissatisfied employees” would change the meaning of the data.

The mean is the sum of numeric scores divided by their count. A mean of 4.07 does not mean 81.4% are satisfied. The mean assumes equal numerical spacing between the categories and can conceal different response distributions. The median is the middle ordered value. Top-box measures intensity of positive responses; favourable share combines the top two categories.

A change from 80% to 85% is **5 percentage points**. The relative percentage increase would be 6.25%, which answers a different question. The notebook uses percentage points for differences between favourable shares.

How to read the charts

- Composition bars count responding records. They describe the sample, not the eligible workforce.
- The 30-item diverging chart shows all five response shares. Half of Sometimes sits on either side of the centre for layout; that does not redefine it as neutral.
- The strengths/opportunities chart compares favourable, Sometimes and unfavourable shares. “Lowest” means lowest within this questionnaire, without an external pass/fail benchmark.
- Boxplots show the median, middle half and spread of block scores. A plotted outlier is not automatically an invalid response.
- Correlation heatmaps show association strength, not an effect of one question on another. The diagonal is 1 because an item is perfectly associated with itself.
- Tenure and department heatmaps show descriptive group means. Read group sizes and the adjusted test table before interpreting a darker or lighter cell as meaningful.
- Comment bars and topic heatmaps show coded mentions. They are not incident rates or a validated sentiment score.

- The topic network connects ideas mentioned in the same answer. It does not connect employees or departments.
- The roadmap chart shows proposed timing relative to approval. It is not evidence that actions have been implemented.

6. Who responded, and why that affects interpretation

Tenure

Category	Records	Share of 327
0 - 3	124	37.9%
3 - 5	53	16.2%
5 - 10	56	17.1%
10 - 15	41	12.5%
15 & Above	53	16.2%

Region

Category	Records	Share of 327
West	171	52.3%
South	73	22.3%
North	54	16.5%
East	24	7.3%
Overseas	5	1.5%

Gender

Category	Records	Share of 327
Male	259	79.2%
Female	67	20.5%
Prefer not to tell	1	0.3%

Department reporting group

Category	Records	Share of 327
Other	65	19.9%
Documentation	59	18.0%
Sales	54	16.5%
Customer Service	51	15.6%
Finance & Accounts	38	11.6%
Operations	36	11.0%
Customs Clearance	13	4.0%
Accounts - Trade	11	3.4%

Mumbai contributes 132/327 records, or 40.4%. Western locations contribute 171/327, or 52.3%. These large shares influence the aggregate result. We do not know whether that reflects the company's actual workforce distribution or uneven participation.

The minimum group size of 10 in the early comparisons is a provisional project reporting rule. Later focused displays use n≥30 and suppress nonzero counts of 1-4. These rules are not guarantees of anonymity or statistical precision. Exact job titles, small locations and unusual combinations can still identify people.

7. What all 30 questions show

The table follows questionnaire order. “Sometimes n” is kept separate from the explicitly unfavourable categories. All rows have valid n=327. The business reading is an interpretation of the observed result, not an additional measured fact.

Credibility

CRE1 Communication from senior leadership is open and honest.

Mean **4.40/5** / Favourable **283/327 (86.5%)** / Sometimes **26** / Unfavourable **18**

Company reading: Protect open communication and examine whether explanations reach employees.

CRE2 Our leaders do what they say they will do.

Mean **4.30/5** / Favourable **270/327 (82.6%)** / Sometimes **39** / Unfavourable **18**

Company reading: Ask where commitments and visible delivery differ.

CRE3 I have confidence in the direction of the organization.

Mean **4.53/5** / Favourable **293/327 (89.6%)** / Sometimes **29** / Unfavourable **5**

Company reading: Confidence in direction is a relative strength to preserve.

CRE4 Senior leaders communicate important changes in a timely manner

Mean **4.34/5** / Favourable **279/327 (85.3%)** / Sometimes **31** / Unfavourable **17**

Company reading: Check whether important changes reach affected people in time.

CRE5 Leaders are willing to listen to employee feedback and act on it.

Mean **4.19/5** / Favourable **261/327 (79.8%)** / Sometimes **39** / Unfavourable **27**

Company reading: Test whether employees see action after providing feedback.

Respect

RES1 My manager genuinely supports my performance, development and well-being.

Mean **4.49/5** / Favourable **288/327 (88.1%)** / Sometimes **23** / Unfavourable **16**

Company reading: Preserve manager support while examining contrasting experiences.

RES2 I receive the resources, support and recognition I need to do my job well.

Mean **4.35/5** / Favourable **271/327 (82.9%)** / Sometimes **39** / Unfavourable **17**

Company reading: Separate resources, support and recognition during follow-up; this item combines them.

RES3 I feel comfortable sharing my views or concerns at work

Mean **4.27/5** / Favourable **270/327 (82.6%)** / Sometimes **32** / Unfavourable **25**

Company reading: Ask what helps employees raise views or concerns; this single item is not a safety diagnosis.

RES4 My efforts and contributions are appreciated and recognised

Mean **4.07/5** / Favourable **237/327 (72.5%)** / Sometimes **56** / Unfavourable **34**

Company reading: Review consistency and usefulness of appreciation and performance feedback.

RES5 I am treated with dignity and respect, regardless of my role or position.

Mean **4.41/5** / Favourable **285/327 (87.2%)** / Sometimes **29** / Unfavourable **13**

Company reading: Preserve respectful treatment and confidentially verify specific concerns.

Fairness

FAI1 Performance, recognition and rewards are fair and based on performance.

Mean **4.07/5** / Favourable **249/327 (76.1%)** / Sometimes **45** / Unfavourable **33**

Company reading: Review performance/reward criteria and explain decisions.

FAI2 Employees have equal opportunities to grow and develop their careers.

Mean **4.24/5** / Favourable **267/327 (81.7%)** / Sometimes **35** / Unfavourable **25**

Company reading: Check access and selection for development opportunities.

FAI3 People-related decisions are made consistently by the Leadership team without favoritism or bias.

Mean **4.12/5** / Favourable **244/327 (74.6%)** / Sometimes **54** / Unfavourable **29**

Company reading: Inspect consistency in applying people-decision criteria.

FAI4 I believe workload and responsibilities are distributed fairly within my team.

Mean **4.31/5** / Favourable **272/327 (83.2%)** / Sometimes **36** / Unfavourable **19**

Company reading: Examine workload allocation, backup and actual work demands.

FAI5 I understand how decisions affecting employees are made.

Mean **4.09/5** / Favourable **247/327 (75.5%)** / Sometimes **56** / Unfavourable **24**

Company reading: Make decision processes and reasons understandable.

Pride

PRI1 I am proud to tell others that I work for Safewater Lines Group

Mean **4.66/5** / Favourable **302/327 (92.4%)** / Sometimes **20** / Unfavourable **5**

Company reading: Preserve company pride through the improvement programme.

PRI2 I understand how my work contributes to the success of the business.

Mean **4.67/5** / Favourable **313/327 (95.7%)** / Sometimes **11** / Unfavourable **3**

Company reading: Strong contribution clarity can support shared workflow goals.

PRI3 I see myself continuing to work here over the next 3 - 5 years.

Mean **4.50/5** / Favourable **286/327 (87.5%)** / Sometimes **23** / Unfavourable **18**

Company reading: Discuss future expectations; intention is not observed retention.

PRI4 I feel that our organization is making a positive impact on our customers and business.

Mean **4.55/5** / Favourable **295/327 (90.2%)** / Sometimes **23** / Unfavourable **9**

Company reading: Preserve the connection employees see with customers and business impact.

PRI5 I feel motivated to contribute beyond my basic job responsibilities when the business needs it.

Mean **4.54/5** / Favourable **295/327 (90.2%)** / Sometimes **24** / Unfavourable **8**

Company reading: Protect willingness to contribute without assuming unlimited extra-work capacity.

Camaraderie

CAM1 People in my team genuinely support and help one another

Mean **4.55/5** / Favourable **298/327 (91.1%)** / Sometimes **20** / Unfavourable **9**

Company reading: Preserve mutual help within teams.

CAM2 Teams across the organization work well together.

Mean **4.29/5** / Favourable **269/327 (82.3%)** / Sometimes **41** / Unfavourable **17**

Company reading: Examine cross-functional handovers and coordination in a pilot.

CAM3 I feel that I belong to this organization

Mean **4.57/5** / Favourable **295/327 (90.2%)** / Sometimes **21** / Unfavourable **11**

Company reading: Preserve belonging while making specific work practices more dependable.

CAM4 I can rely on colleagues from other teams when I need their support.

Mean **4.23/5** / Favourable **262/327 (80.1%)** / Sometimes **46** / Unfavourable **19**

Company reading: Clarify how to obtain support from other functions.

CAM5 People here celebrate each other's successes and achievements.

Mean **4.13/5** / Favourable **238/327 (72.8%)** / Sometimes **68** / Unfavourable **21**

Company reading: Ask how employees experience shared acknowledgement of achievements.

Overall employee experience

OVR1 I would recommend this organization as a great place to work.

Mean **4.52/5** / Favourable **292/327 (89.3%)** / Sometimes **25** / Unfavourable **10**

Company reading: Advocacy is positive overall; the item does not calculate standard eNPS.

OVR2 I have opportunities to learn, grow and build my career here.

Mean **4.49/5** / Favourable **296/327 (90.5%)** / Sometimes **16** / Unfavourable **15**

Company reading: Preserve learning opportunities and examine access/progression concerns.

OVR3 I have what I need to perform at my best and contribute to business success.

Mean **4.50/5** / Favourable **297/327 (90.8%)** / Sometimes **23** / Unfavourable **7**

Company reading: Verify specific resource gaps even with positive overall answers.

OVR4 I am satisfied with the overall employee experience at Safewater Lines.

Mean **4.39/5** / Favourable **285/327 (87.2%)** / Sometimes **26** / Unfavourable **16**

Company reading: Overall satisfaction can coexist with lower ratings on specific experiences.

OVR5 I believe Safewater Lines provides a workplace where I can do meaningful and fulfilling work.

Mean **4.51/5** / Favourable **292/327 (89.3%)** / Sometimes **25** / Unfavourable **10**

Company reading: Meaningful work is a strength and an exploratory association topic.

The six supplied blocks

Block	Mean /5	Pooled favourable %	Alpha
CRE	4.35	84.77%	0.88
RES	4.32	82.63%	0.88
FAI	4.17	78.23%	0.86
PRI	4.58	91.19%	0.84
CAM	4.35	83.30%	0.79
OVR	4.48	89.42%	0.90



Figure 2. Credibility, Respect and Fairness. n=327 valid responses per item. Grey means Sometimes; red means 1-2. Right-hand labels report the favourable 4-5 share.

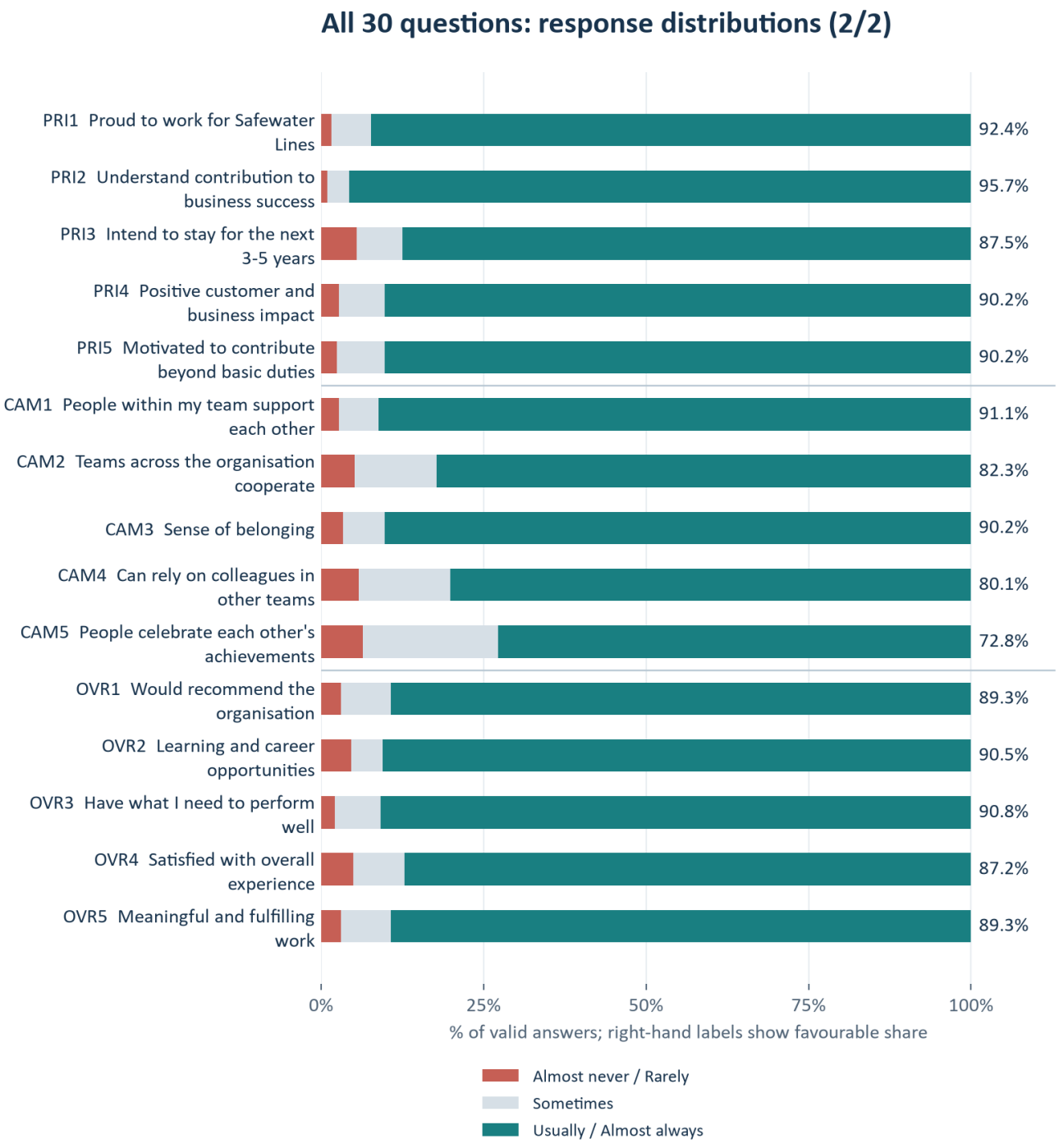


Figure 3. Pride, Camaraderie and Overall Experience. These are separate questions; an intention-to-stay answer is not observed retention.

Pride is highest and Fairness lowest on both working mean and pooled favourable share. The overall culture mean is **4.354/5**. Across all 30 questions, **84.9% of item answers** are favourable. That is an answer-level summary, not a claim that 84.9% of employees are satisfied. Satisfaction itself is OVR4: **285/327, or 87.2%**, favourable.

The mean and favourable share can rank items differently. RES4 and FAI1 both have a mean of about 4.07, but their favourable shares are 72.5% and 76.1%. Different mixes of 1-5 answers explain that. The notebook correctly retains both summaries and uses favourable share as the main presentation ranking.

What deserves preservation

Contribution clarity PRI2 is 95.7% favourable, company pride PRI1 92.4%, and within-team support CAM1 91.1%. The preserve prompt also identifies valued team/colleague relationships in 119 records, support/respect in 77, culture/belonging in 75 and trust/autonomy in 61. These categories overlap.

The useful company inference is that the improvement programme can build on shared purpose and working relationships. Track those strengths during process changes, because an intervention that improves documentation while making support less accessible would need adjustment.

What deserves closer attention

Recognition RES4 is lowest at 72.5%, followed by celebrating achievements CAM5 at 72.8%. Consistency of people decisions FAI3 is 74.6%, understanding decisions FAI5 75.5%, and fair performance/reward decisions FAI1 76.1%. This gives managers specific questions to examine rather than one vague instruction to “improve culture.”

8. Reliability, correlations and what they can tell us

Cronbach's alpha

Alpha checks how strongly questions in a proposed block vary together. The notebook calculates it as:

```
alpha = k/(k-1) × [1 - sum(item variances)/variance(sum of items)]
```

Here k=5 for each block. The block alphas range from approximately 0.79 to 0.90; all 30 items together have alpha **0.9665**. These are consistency diagnostics under assumptions. They do not prove that the six named blocks are distinct, that the questionnaire measures the intended concepts, or that an overall score is valid for predicting an outcome. Adding many correlated items can increase alpha.

Corrected item-total correlations compare an item with the sum of the other four questions in its block, excluding the item itself. This avoids a mechanical self-correlation. The implementation uses ordinary Pearson correlation for this diagnostic. A lower value invites a wording/construct review; it does not automatically justify deleting a useful question.

Spearman associations

Spearman correlation asks whether records giving higher answers to one item also tend to give higher answers to another. It uses ranks and ranges from -1 to +1. The notebook's average within-block item association is **0.559**, compared with **0.494** across all item pairs. Items overlap substantially; these two averages do not determine how many underlying factors exist.

Three outcomes are kept separate: recommendation OVR1, intention to stay PRI3 and satisfaction OVR4. All three are excluded from the predictor list. This avoids direct self-prediction, but it does not remove shared response style, overlapping wording or confounding.

Outcome	Item	Related question	Rho
recommend	OVR5	I believe Safewater Lines provides a workplace where I can do meaningful and fulfilling work.	0.677
recommend	OVR2	I have opportunities to learn, grow and build my career here.	0.635
recommend	FAI2	Employees have equal opportunities to grow and develop their careers.	0.607
stay 3-5 yrs	OVR2	I have opportunities to learn, grow and build my career here.	0.572
stay 3-5 yrs	OVR5	I believe Safewater Lines provides a workplace where I can do meaningful and fulfilling work.	0.528
stay 3-5 yrs	PRI1	I am proud to tell others that I work for Safewater Lines Group	0.523
satisfied	OVR5	I believe Safewater Lines provides a workplace where I can do meaningful and fulfilling work.	0.672
satisfied	OVR2	I have opportunities to learn, grow and build my career here.	0.636
satisfied	FAI2	Employees have equal opportunities to grow and develop their careers.	0.617

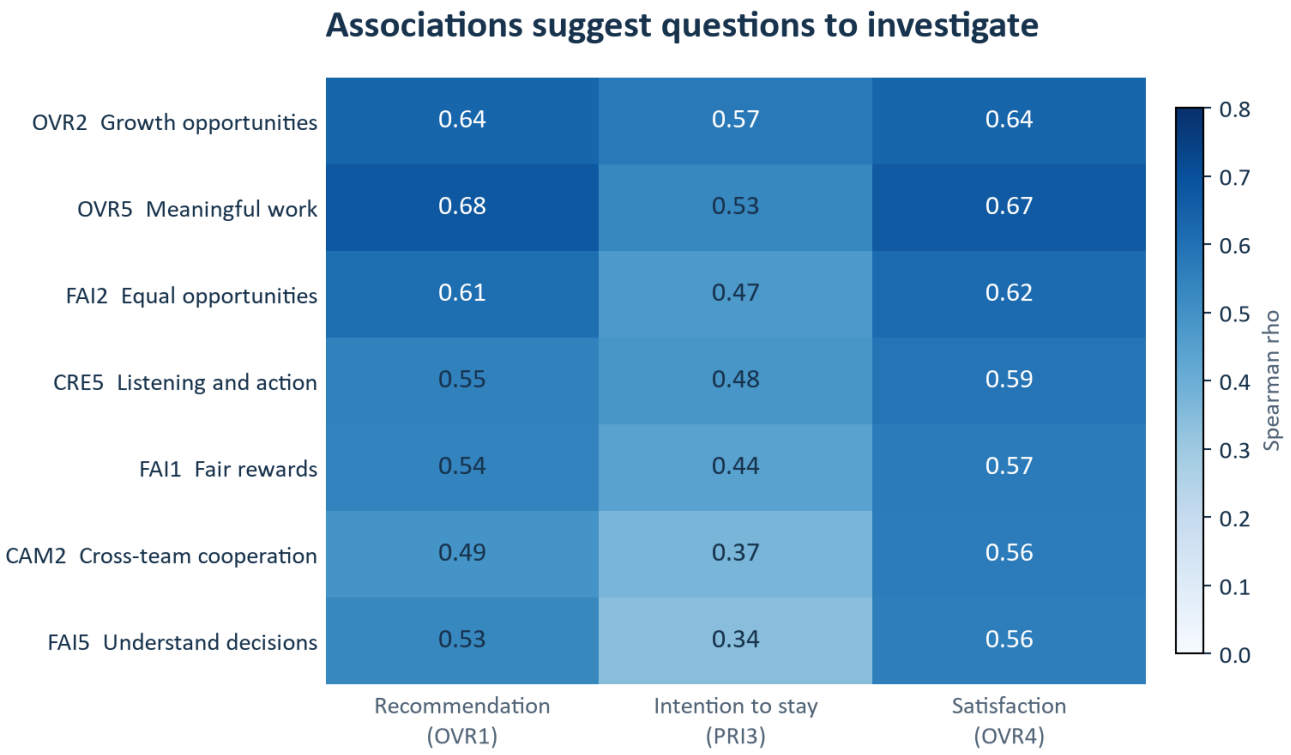


Figure 4. Selected exploratory item-outcome associations, recomputed from the processed ratings. This is not a causal driver ranking or predicted effect of an intervention.

For the company, growth, meaningful work, equal opportunities and listening merit useful follow-up conversations. The correlations do not rank the return from investing in each topic. For example, rho=0.572 between growth opportunity and intention to stay is not a 57.2% increase in retention, and it does not tell us what would happen after a training programme.

The notebook does not calculate standard eNPS from OVR1. That item uses this survey's five-point frequency scale. PRI3 is an intention at the time of the survey; actual retention and future departures were not observed.

9. Group comparisons and response-style sensitivity

Tenure

Tenure	n	Culture mean	Growth mean	Stay-intention mean
0 - 3	124	4.30	4.40	4.41
3 - 5	53	4.21	4.30	4.25
5 - 10	56	4.38	4.59	4.61
10 - 15	41	4.70	4.83	4.90
15 & Above	53	4.32	4.49	4.51

Tenure	Recognition favourable	Growth favourable
0 - 3	73.4%	87.9%
3 - 5	66.0%	84.9%
5 - 10	73.2%	94.6%
10 - 15	85.4%	97.6%
15 & Above	66.0%	92.5%

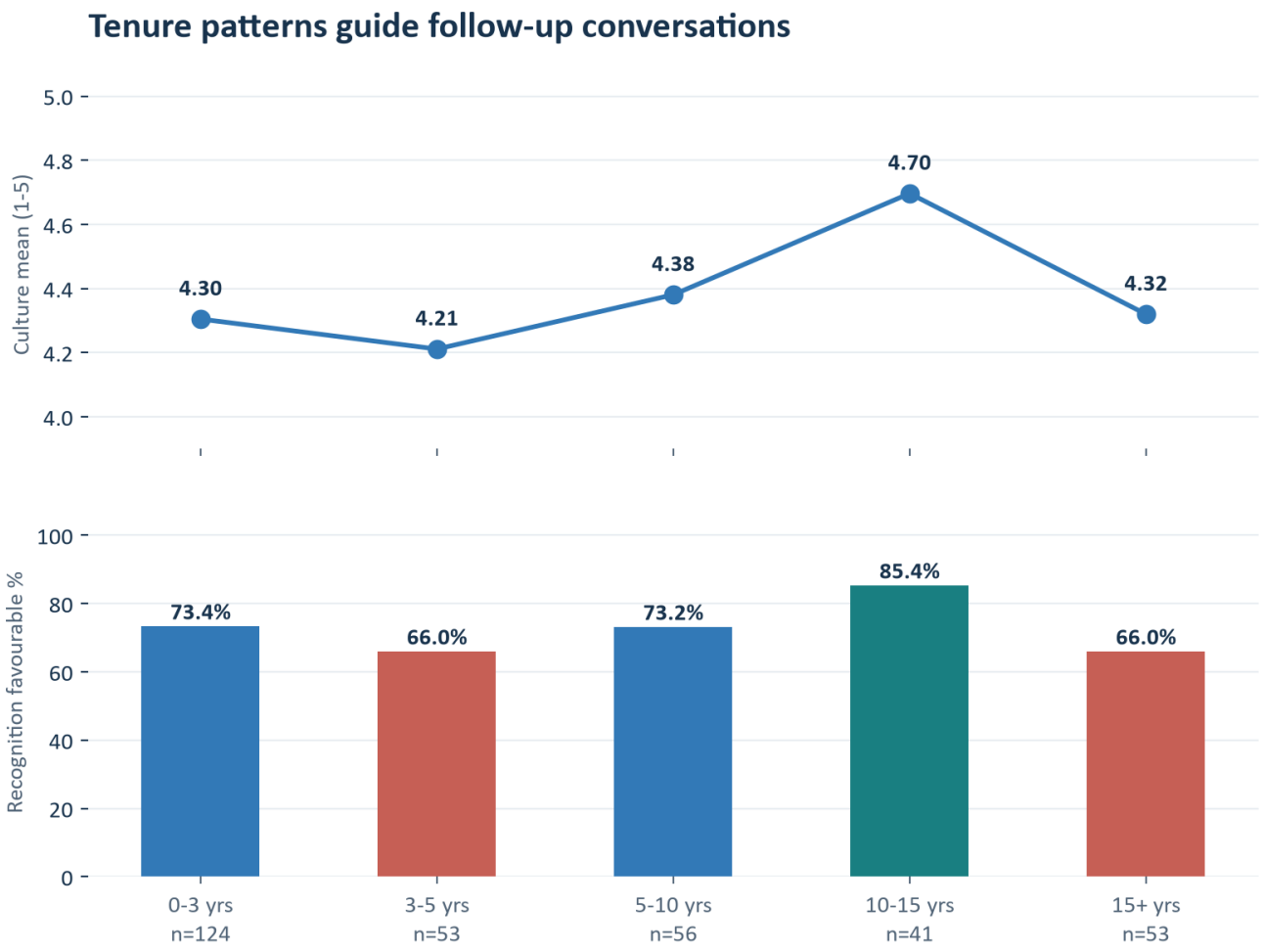


Figure 5. The same survey compares different tenure groups at one time. Lower scores in one group guide follow-up; they do not establish a tenure effect or predict departure.

The 3-5-year group has the lowest observed culture mean, growth mean and intention-to-stay mean in this set of tenure groups. The 10-15-year group has the highest culture mean. Recognition favourability is 66.0% in both the 3-5-year and 15+ groups, so the recognition issue is not exclusive to mid-tenure employees.

A practical follow-up is to ask what employees expect after several years: progression criteria, useful feedback, role growth, recognition and access to opportunities. The data do not prove that tenure causes lower scores or identify who will leave. Different roles, departments and employee selection may explain part of the pattern. Comparing different tenure groups at one time is not the same as following the same employees as they age in the company.

What the tests add

The notebook uses Kruskal-Wallis rank comparisons for eligible independent groups, then jointly adjusts **11** full-sample/sensitivity tests with Benjamini-Yekutieli, or BY. The adjustment addresses multiple testing with overlapping tests. It does not correct nonresponse or establish business importance.

Sample	Comparison	Eligible n	Original p	BY-adjusted p
All records	tenure	327	0.0012	0.0279
All records	level	318	0.2375	0.7172
All records	region	322	0.0378	0.2509
All records	dept_group	327	0.0113	0.0939
All records	gender	326	0.0837	0.3475
All records	gave_name	327	0.0053	0.0583
Excluding identical-category records	tenure	273	0.0017	0.0279
Excluding identical-category records	level	258	0.0613	0.2910
Excluding identical-category records	region	268	0.0516	0.2858
Excluding identical-category records	dept_group	273	0.1460	0.4849
Excluding identical-category records	gender	272	0.1263	0.4661

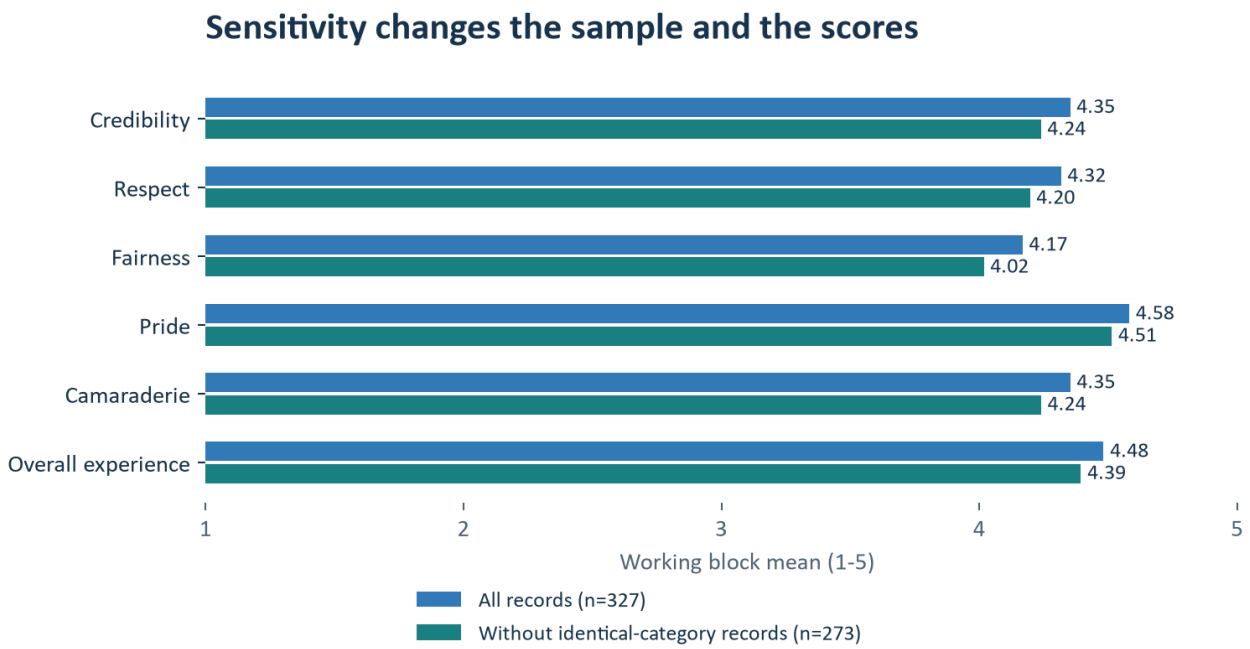


Figure 6. The primary sample retains all records. Excluding identical-category records is a sensitivity comparison, not a correction toward a known true score.

At an exploratory adjusted threshold of 0.05, tenure remains a signal in both full and sensitivity samples, with adjusted p about **0.0279**. The listed department, region, role, gender and name-supply comparisons do not meet that adjusted threshold. This does not prove equality, nor does the tenure omnibus result identify which particular pair differs. Appropriate pairwise evidence would be needed for that claim.

The departmental patterns can still guide listening. For example, Customer Service has an observed culture mean of 4.09 among 51 records. It has 29 records with at least one fairness item at 1-3 and 10 with the within-team-favourable/cross-team-1-3 pattern. These are reasons to understand its work context, not evidence that Customer Service causes the organisational problem. A department can be experiencing the consequences of a workflow problem elsewhere.

Straight-liners

There are **54 records, or 16.5%**, with the same rating on all 30 questions: 51 use 5 throughout, two use 4 and one uses 3. Identical answers may reflect consistent perceptions or response style. We have no independent basis to discard them as dishonest or careless.

The primary analysis keeps them. Excluding them leaves n=273 and reduces every block mean: Fairness falls from **4.168 to 4.018**, and Pride from **4.582 to 4.514**. Recognition favourable share falls from **72.5% to 67.4%**. The same main strengths and lower-scoring items remain visible. This establishes sensitivity to a sample choice; it does not reveal a corrected true score.

The named group has a higher observed culture mean than the unnamed group, approximately 4.42 versus 4.20. That difference could reflect who chose to name themselves and their work circumstances. The data do not establish that name disclosure caused more positive answers, or that all ratings are inflated by a known amount.

10. The single-choice HR priority

Mapped selected HR theme	n	% of 327
Communication	80	24.5%
Grievances	64	19.6%
Development	51	15.6%
Benefits & Facilities	51	15.6%
R&R	30	9.2%
Process & Policies	20	6.1%
Approachable	18	5.5%
Engagement	13	4.0%

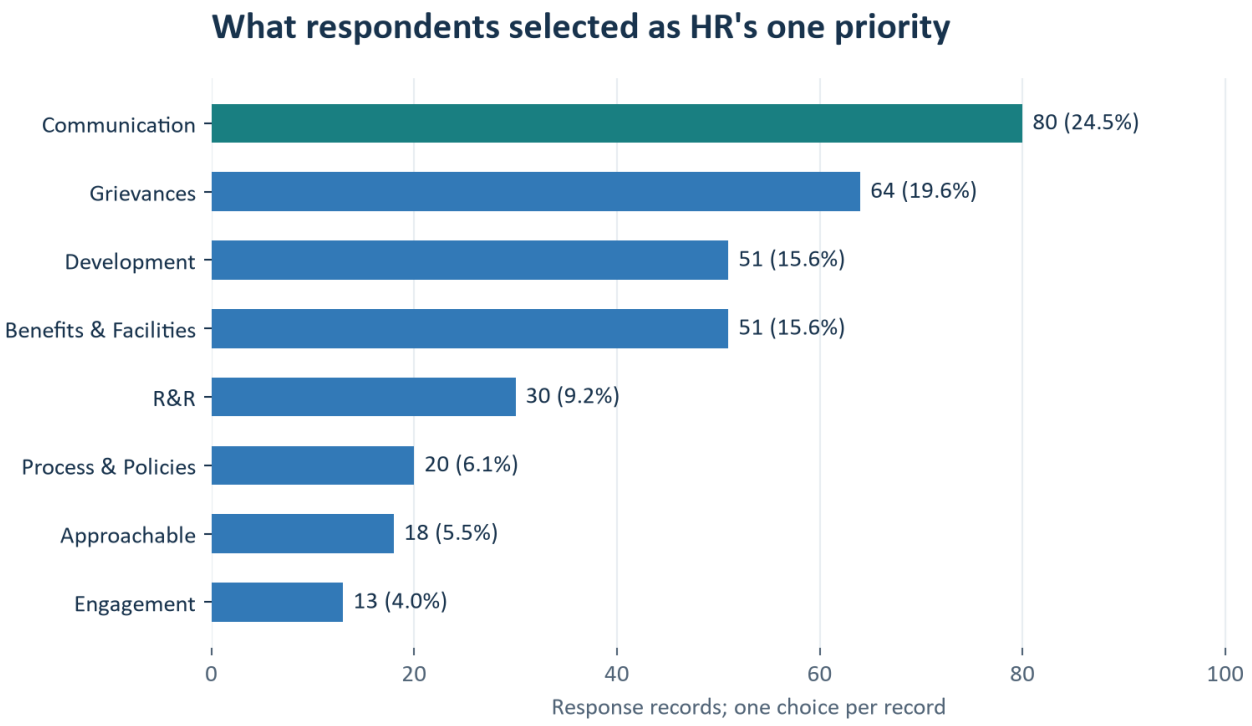


Figure 7. One selected priority per record. Communication combines listening (67) and keeping employees informed (13); their different meanings matter.

The most selected individual option is **listening to employees and their concerns, 67 records**. The mapped Communication category combines those 67 with **13 choosing keeping employees informed**, giving 80/327 or 24.5%.

This distinction matters: the evidence supports examining listening and response, as well as information delivery. Simply increasing announcement frequency may leave the largest component of this theme unanswered. Grievances combines resolving problems and supporting employees. Each respondent chose only one option; the table does not describe every need they have.

Cramér's V describes association between demographic categories and HR themes. The notebook also examines expected cell frequencies. All five examined tables contain sparse expected cells, so it withholds routine chi-square p-values. Descriptive V values alone do not establish a reliable departmental difference in HR needs.

11. How written answers become themes

Coverage and the coding process

There are **1,892 nonblank answers from 321 response records**. The 327×6=1,962 possible comment cells therefore include 70 blanks. A short response can still be meaningful, and a nonblank response can still be uninterpretable or a refusal to answer.

The codebook contains 15 broad topics. Each topic has inclusion rules, exclusions and a paraphrased illustration. Each nonblank answer has an explicit ledger entry linking its question and source-row position to one or more topic codes and a stance.

This is **applied, codebook-based content analysis**. The semantic decisions were made by an assistant and stored in the notebook. They are not generated by a keyword classifier each time the code runs. A rerun reproduces assignments and counts; it does not reread the text and independently validate every interpretation.

The stance separates a suggestion/concern, current strength, mixed account, no concern/no change, nonresponse, unclear reply, unresolved reference and no valued aspect. Thus a no-concern fairness reply is retained instead of being counted as a complaint or removed as empty text.

The ledger markers encode those distinctions: `+` indicates a current strength, `~` a mixed account, `!` a suggestion/concern, `=` no concern/change, `-` a nonresponse, `?` unclear, `0` no valued aspect and `>` an unresolved reference. With no explicit marker, the preserve question defaults to strength and the improvement questions to suggestion/concern. The marker is an encoding convention; it is not an automatic sentiment judgment.

The text/order hash prevents old row-position labels from being silently attached to changed comments. It verifies a serialized text snapshot, not the accuracy of the interpretation.

How counts avoid double-counting

For request counts, the notebook includes suggestion/concern and mixed answers across all questions, then counts each source record once per topic. If the same record asks for communication improvements in three answers, it contributes one communication requester.

For strengths, it counts explicit valued topics in the preserve question. A person can value their team and request better teamwork at the same time. These are different contexts, so subtracting strength counts from request counts would have no defined meaning.

Code	Topic	Request n	Request %	Preserve n
c	Communication and decision explanations	194	59.3%	12
r	Recognition and performance feedback	179	54.7%	4
t	Teamwork and cross-team coordination	176	53.8%	119
m	Support and respectful treatment	139	42.5%	77
d	Career growth and learning	136	41.6%	18
v	Employee voice and follow-through	134	41.0%	10
f	Fair treatment and decision consistency	114	34.9%	3
p	Role clarity and work processes	91	27.8%	7
w	Workload, boundaries and flexibility	78	23.9%	6
b	Pay, incentives and benefits	51	15.6%	9
i	Trust, integrity and autonomy	49	15.0%	61
s	Facilities, tools and staffing	39	11.9%	0
e	Activities and social connection	38	11.6%	4
o	Business identity and customer service	10	3.1%	22
k	Work culture and belonging	8	2.4%	75

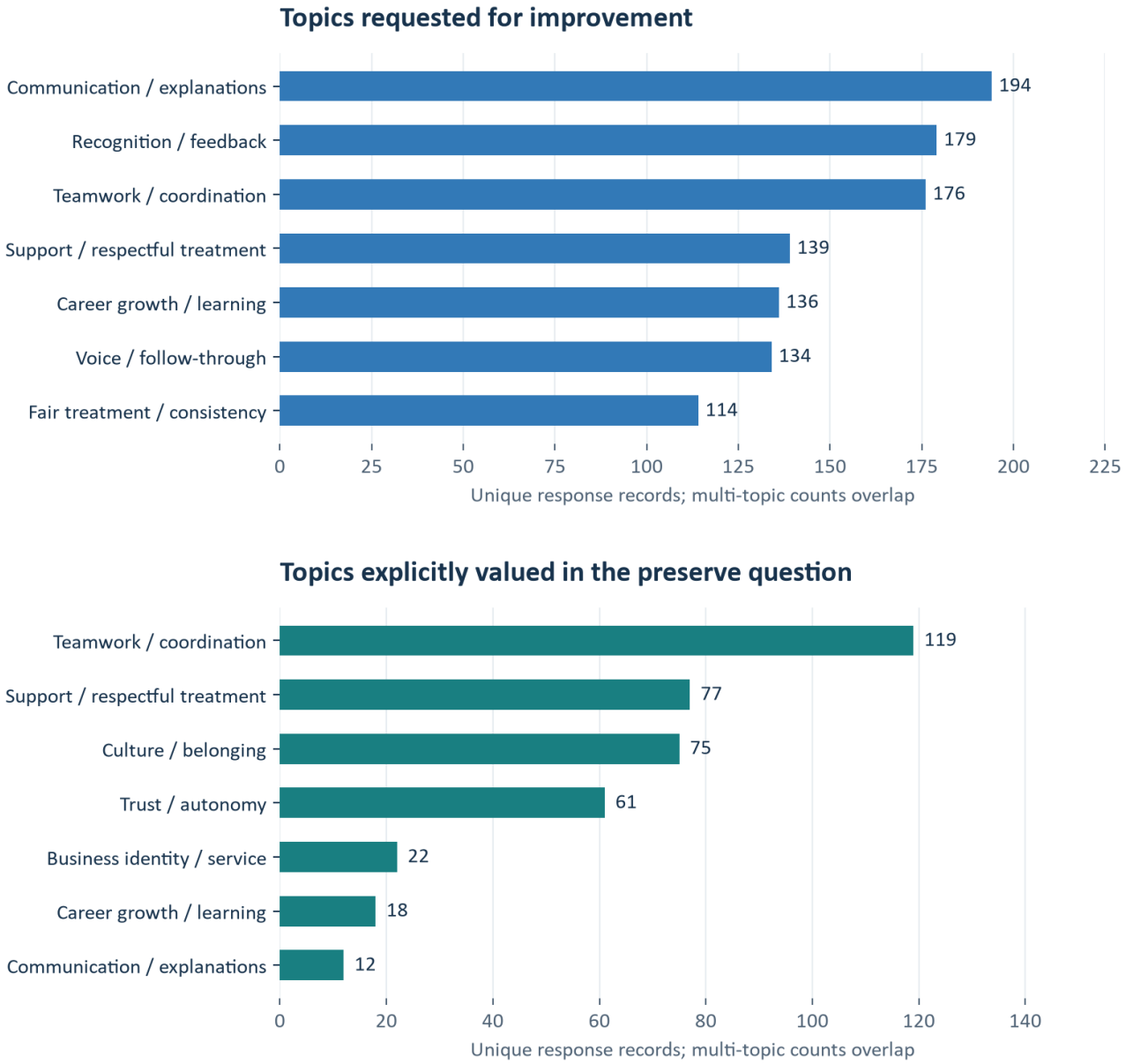


Figure 8. Requests and strengths use different prompt/stance rules. Count each response record once per topic. Topics overlap and counts are not severity or incident rates.

Percentages above use all 327 source records. An alternative base of 321 commenters is also available in the notebook. The 15 percentages do not sum to 100% because an answer and a record can contain several topics. These are prompted mentions, not workforce prevalence, incident rates or severity scores.

Communication is broad: it includes information sharing and decision explanations. Recognition includes constructive performance feedback as well as appreciation and awards. The teamwork topic includes both relationships and coordination. Managers should read the code definitions before turning a broad label into a narrow intervention.

What the review achieved

The same assistant reread **568 distinct comments**, including all 104 original flags and specified topic cohorts. It changed **53 assignments/statuses**. Career/learning requester counts changed from 143 to 136 as overinterpretation of unspecified opportunities and company growth was removed. Recognition remains 179; team/colleague strengths remain 119.

There are currently **1,590 comments with at least one topic**, while the ledger accounts for all 1,892 nonblank answers. The remaining 302 include no-concern, nonresponse, unclear and other answers without assigned topics. Lack of a topic does not necessarily mean lack of a useful response. **75 cases still need contextual clarification or language/reference confirmation**.

This is not independent human validation, an accuracy estimate or an inter-rater reliability result. A teammate should review material that supports the proposed priorities, uncertain language and accounts requiring confidential verification before final external presentation.

Bigrams and trigrams

These are exact adjacent two-word and three-word sequences. They help locate repeated wording. The code preserves token adjacency and does not create artificial phrases by first deleting internal stopwords. It separates sentences/comments/questions, counts one phrase occurrence per record within a question and displays phrases supported by at least five records.

Names and certain titles are filtered before export. The extraction is English-focused, so spelling, paraphrases and non-English/transliterated answers may be missed. Phrase counts support navigation; they do not replace contextual coding or prove that repeated text came from independent experiences. The code's number-filtering intention also needs care: numeric tokens are omitted by the tokenizer, so a window can join alphabetic words separated by a number. That is a method limitation to remember when calling every extracted phrase literally adjacent in the original text.

12. The two principal organisational questions

A. Cross-team execution

Within-team support CAM1 is **298/327 = 91.1% favourable**. Cross-team collaboration CAM2 is **269/327 = 82.3%**. Access to colleagues in other teams CAM4 is **262/327 = 80.1%**. The 8.9 percentage-point CAM1-CAM2 gap compares different aspects of cooperation.

The paired table is more informative than comparing two percentages in isolation:

Same response records	Cross-team favourable	Cross-team 1-3	Total
Within-team favourable	253	45	298
Within-team 1-3	16	13	29
Total	269	58	327

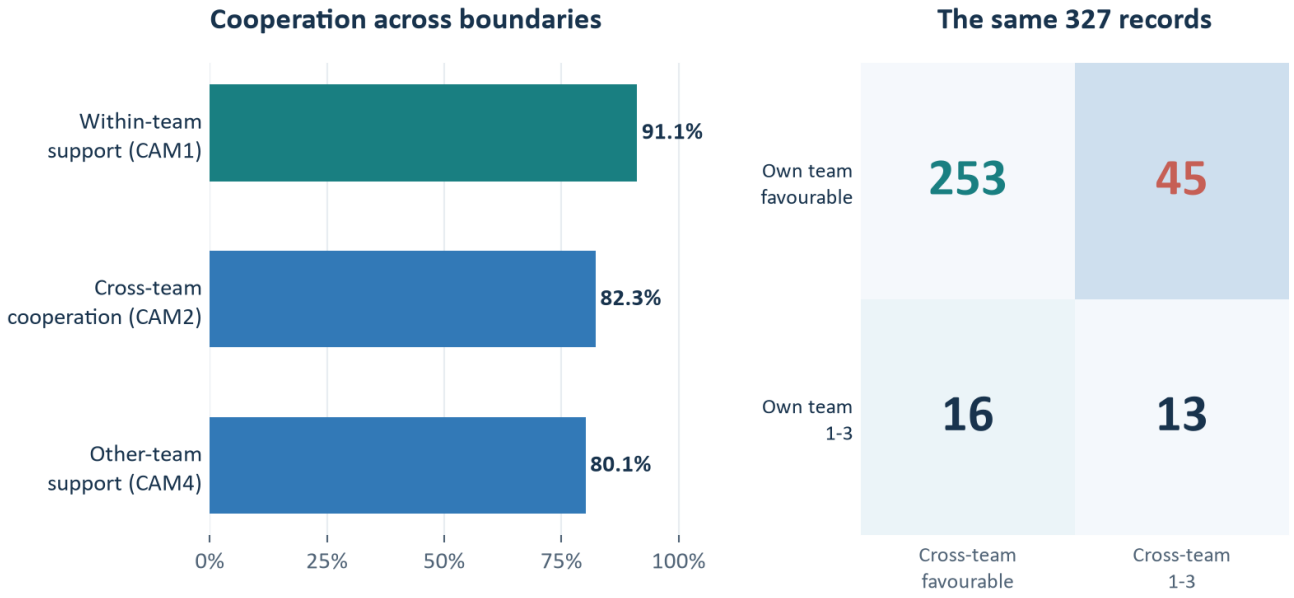


Figure 9. The paired table compares the same records. 45 rate within-team support favourably but cross-team cooperation 1-3; 16 show the reverse. 1-3 includes Sometimes.

The 45-versus-16 discordance shows that more records give favourable within-team support alongside a lower cross-team answer than the reverse. Of those 45, **34 answer Sometimes and 11 answer Almost never/Rarely** on cross-team collaboration. The core opportunity therefore includes consistency of cooperation as well as more clearly negative experiences.

Across the original five categories, 90 records rate within-team support higher, 203 give the same category and 34 rate it lower. Equal original categories do not necessarily mean both answers are favourable.

In the teamwork prompt, 124 answers request communication-related changes and 53 discuss roles/processes; 32 discuss activities. Topics overlap and are not a vote between mutually exclusive programmes. They support investigating information requirements, receiving ownership, backup, response expectations and escalation in a real workflow.

Useful inference: employees can have helpful immediate colleagues while encountering uncertainty when work crosses functions. A small handover pilot is a plausible test. The survey does not tell us which specific department pair, handover or system causes delay, nor does it measure actual rework or customer impact.

Keeping the count of 45 after excluding straight-liners is not an independent robustness result: a record using the same category for every item cannot be in that discordant group by definition.

B. Fairness of people decisions

The five fairness questions should be read separately:

Item	Favourable n (%)	Sometimes n	Unfavourable n
FAI1	249 (76.1%)	45	33
FAI2	267 (81.7%)	35	25
FAI3	244 (74.6%)	54	29
FAI4	272 (83.2%)	36	19
FAI5	247 (75.5%)	56	24

FAI1: Performance, recognition and rewards are fair and based on performance.

FAI2: Employees have equal opportunities to grow and develop their careers.

FAI3: People-related decisions are made consistently by the Leadership team without favoritism or bias.

FAI4: I believe workload and responsibilities are distributed fairly within my team.

FAI5: I understand how decisions affecting employees are made.

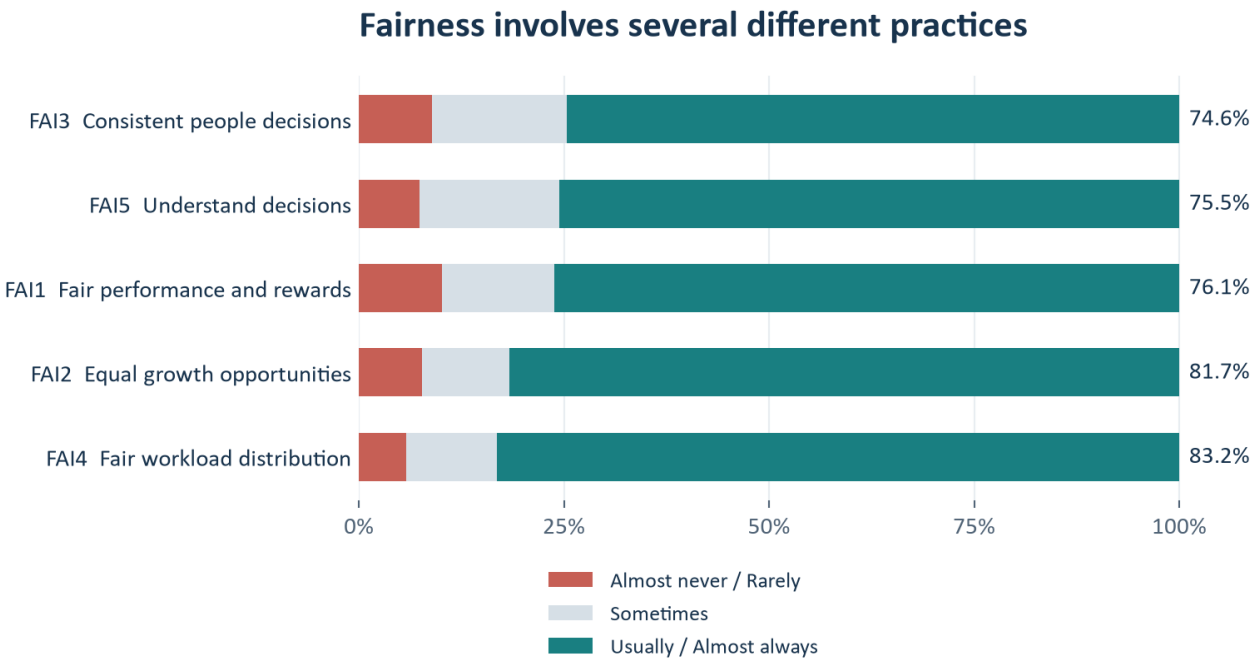


Figure 10. Fairness covers decision consistency, explanations, rewards, opportunity and workload. The five questions point to different practical investigations.

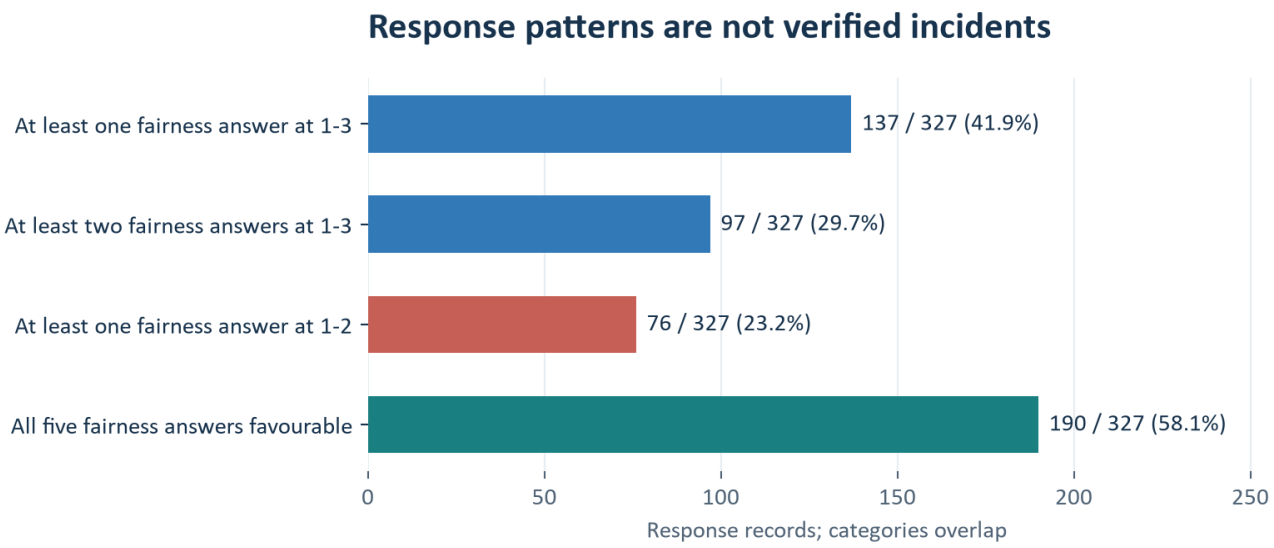


Figure 11. These are overlapping descriptive response patterns, not verified unfair-treatment prevalence. In particular, 1-3 includes Sometimes.

The relatively lower decision-consistency, explanation and reward items suggest examining how criteria are applied and understood. Workload and opportunity also matter, but a single generic fairness programme would obscure these different issues.

137/327 records, or 41.9%, have at least one fairness item at 1-3; **97** have at least two. **76 records, or 23.2%,** have at least one explicitly unfavourable fairness answer at 1-2. **190 records** rate all five fairness items favourably. None of these patterns by itself verifies unfair treatment or discrimination.

The fairness comments include 67 career/learning requests, 66 recognition requests, 64 communication/explanation requests and 36 workload requests, with overlap. They also include **73 no-concern replies**, or 23.7% of the 308 nonblank fairness answers. Retaining those replies is necessary to show mixed experiences accurately.

Cross-team and fairness patterns overlap but remain distinct. Of 58 records giving CAM2 a 1-3 response, 47 also have a fairness item at 1-3. Another **90 records** have a fairness item at 1-3 despite favourable CAM2. Improving handovers alone is therefore not a complete response to the fairness pattern.

Useful inference: test accessible appraisal/progression/reward criteria, reasons for decisions, a review route and whether comparable decisions use comparable criteria. A published policy is an implementation output; employee understanding and perceived consistency are separate outcomes.

13. Additional insights from checking the same records

The following are additional descriptive cross-checks for this walkthrough. They are not causal tests or new validated measures.

Strong pride can coexist with inconsistent appreciation

73 records rate company pride favourably and recognition at 1-3. **79 records** rate contribution clarity favourably and recognition at 1-3. **58 records** rate overall satisfaction favourably and recognition at 1-3.

This tells the company that an employee can understand the purpose of their work, value the organisation and still want more reliable appreciation. A favourable overall satisfaction item should not close the discussion about specific daily experiences. Ask managers to make feedback specific, timely and linked to explainable criteria, then ask whether employees find it useful.

Also, **77 of the 90 records** with recognition at 1-3 have at least one fairness item at 1-3. Recognition and fair decisions may need coordinated follow-up. The overlap does not establish which causes which.

Growth availability and equal access are different questions

Overall growth opportunity OVR2 is 90.5% favourable, while equal opportunity FAI2 is 81.7%. **36 records** rate their growth opportunity favourably while rating equal opportunity at 1-3; only **7** show the reverse pattern.

The company can investigate who can access opportunities, how selection works and whether progression criteria are understandable, alongside whether learning opportunities exist. Simply adding more courses may not answer an access or consistency concern. The survey does not demonstrate discriminatory allocation or verify individual promotion cases.

Communication quality includes whether something happens next

Leadership honesty CRE1 is 86.5% favourable; listening and acting CRE5 is 79.8%. **39 records** rate honesty favourably and listening/action at 1-3; **17** show the reverse. In the HR question, listening attracts 67 selections, while keeping people informed attracts 13.

These patterns support a concern-to-action process with acknowledgement, ownership, updates and closure feedback. The inference is a testable opportunity to improve follow-through. Neither increased message volume nor an issue marked closed proves that employees felt heard.

Recognition is an observable management practice to test

RES4 and CAM5 are the two lowest favourable shares, and 179 records request recognition/performance feedback. Of those requesters, **126 still rate RES4 favourably**. A request may mean “more consistent,” “more specific” or “fairer access,” rather than “this never happens.” Ask about the kind and usefulness of recognition before choosing awards, events or monetary changes.

Specific accounts need a separate verification route

The comments contain accounts/requests concerning facilities or equipment, respectful correction and time boundaries. Broad coded topic totals cannot establish counts of serious incidents. Confidentially verify the particular conditions and route confirmed issues through the company's existing processes. This work can proceed alongside broad process pilots without publishing names or assuming an allegation is established fact.

14. What the topic network adds

The primary network uses **1,230 topic-coded suggestion/mixed answers from 305 response records**. Thirteen topics meet the primary display threshold of at least 20 included answers. The codebook itself still has 15 topics. Links are shown when at least 10 answers mention both topics.

Topics discussed together in the same answer

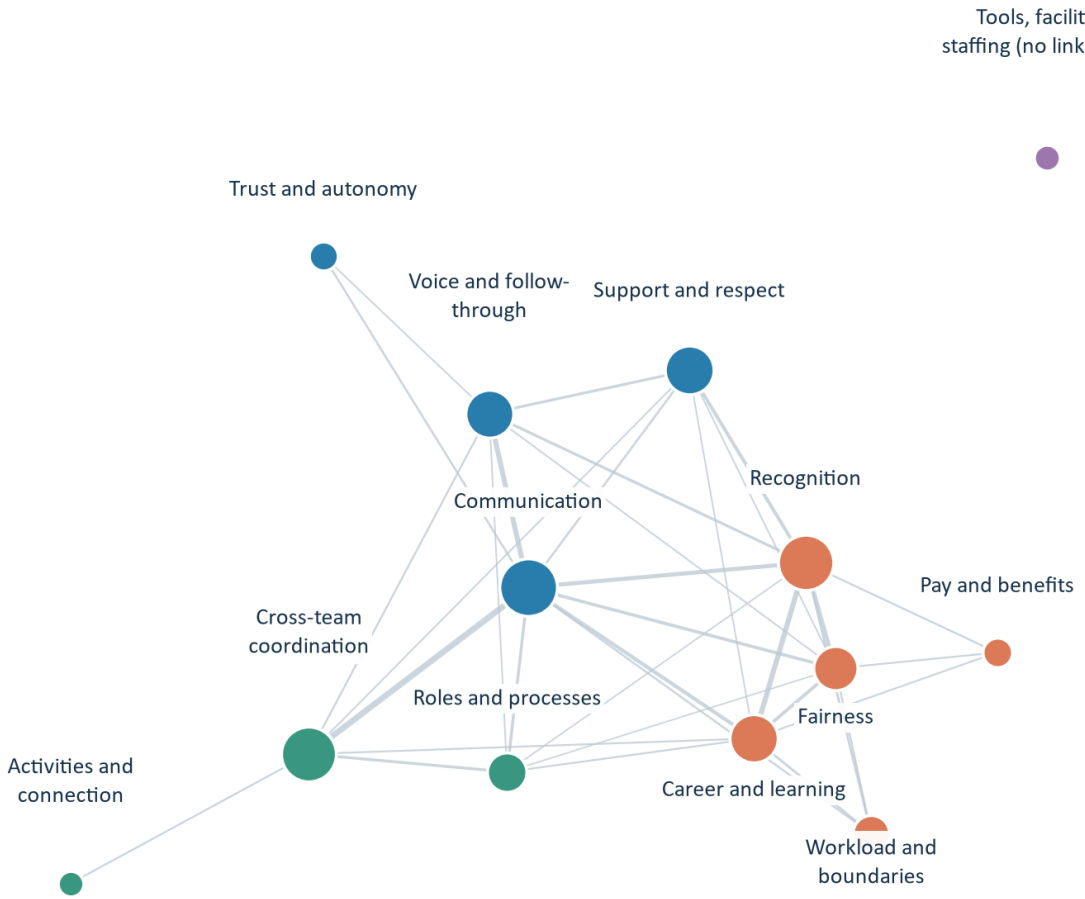


Figure 12. Node area represents unique records and links count same-answer topic pairs with support of at least 10. Colours/layout are custom exploratory choices. The network is strongly prompt-dependent.

For a single answer with three topic labels, the code counts three pairs, once each. A record can supply several answers, so same-answer pair counts and unique contributor counts are separate. Merely mentioning two topics in separate answers from the same record does not create a link.

The association weight is `c_ij / (c_i × c_j)`, where c_ij is pair support and c_i/c_j are topic supports in included answers. The interface also shows this times the included answer count, labelled lift. It adjusts for topic frequency but is not a significance test, effect size or severity measure.

Node area represents unique record support. Link width represents same-answer co-mentions. Colours come from a local greedy weighted-modularity grouping. A local layout balances weighted attraction, repulsion and weak confinement; it tries four seeded initializations and rotates the result for readability. Axes have no substantive meaning. Isolated topics are placed separately, so their position should not be interpreted as a measured psychological distance.

The implementation is VOSviewer-inspired, with compatible import files. It is not a native VOSviewer run and does not identify employee communities, validated factors or causal pathways.

Prompt dependence is material. Communication and teamwork co-occur in 98 answers from 93 records, but **85 of those answers come from the teamwork prompt**. Excluding that prompt leaves 13. Fairness and career co-occur in 46 answers, of which 41 come from the fairness question; excluding it leaves five. Therefore the map helps organise related discussion topics but does not independently validate the headline story.

The interactive controls exclude each question and change thresholds. The notebook checks four link thresholds and three clustering resolutions. Changes in clusters demonstrate dependence on choices. Fixed positions and colours in exclusion views support visual comparison; they are not a rerun of the clustering for every view.

15. Why the notebook does not need every advanced method

The fuzzy check

The notebook calculates each item's observed response-frequency profile, averages the profiles within a block and multiplies by scores 1-5. With complete data and equal item weights, that result is exactly the ordinary mean. It verifies equality to numerical precision. Reporting it as an additional fuzzy insight would repeat an existing statistic.

That particular check does not reject every fuzzy method. A substantive fuzzy model would require a defined decision, justified membership functions/weights, validation and sensitivity analysis. Those inputs are not supplied. Fuzzy membership is not automatically probability or uncertainty in an employee's answer.

Other conditional methods

Factor analysis could address a necessary question about underlying measurement structure, but the current summaries are explicitly descriptive. Regression could model conditional associations, but it would not establish causality from this one-time survey. Attrition prediction would need observed departure outcomes and validation. Generic sentiment or a word cloud would lose the question-specific meaning of short replies. None of these methods should be added solely to make the project look more technical.

16. What the company can do with the findings

The notebook organises the proposed programme around two primary priorities and three supporting themes. This prioritisation was selected in the documented project work; the descriptive evidence supports examining it, but an algorithm did not prove it to be the only possible order.

Cross-team execution

Decision detail	Proposal / evidence
Evidence	CAM1 91.1% vs CAM2 82.3%; paired 45 vs 16
Pilot action	Map one workflow; define information, recipient, backup, acknowledgement and escalation
Accountable role	Operations/functional head
What to measure	Handover completeness, agreed timeliness, rework, CAM2/CAM4

Fairness of people decisions

Decision detail	Proposal / evidence
Evidence	FAI3 74.6%, FAI5 75.5%, FAI1 76.1%; mixed comment experiences
Pilot action	Clarify criteria, reasons and review access; inspect application to a defined decision set
Accountable role	HR head with business leadership
What to measure	Criteria access, explanation coverage, FAI1/FAI3/FAI5

Recognition and feedback

Decision detail	Proposal / evidence
Evidence	RES4 72.5%; CAM5 72.8%; 179 requester records
Pilot action	Offer useful specific feedback and explain recognition criteria across pilot roles
Accountable role	People managers with HR
What to measure	Access, usefulness and fairness of feedback; RES4/CAM5

Roles, workload and backup

Decision detail	Proposal / evidence
Evidence	91 role/process requests and 78 workload/boundary requests
Pilot action	Define critical-task owners, cover and escalation expectations
Accountable role	Functional head
What to measure	Backup coverage, verified workload problems, FAI4

Voice and visible action

Decision detail	Proposal / evidence
Evidence	CRE5 79.8%; 67 choose listening as HR priority
Pilot action	Acknowledge concerns, assign an owner/update date and seek closure feedback
Accountable role	HR/leadership with issue owners
What to measure	Acknowledgement, ownership, useful resolution, CRE5

The company can use existing tools for a small pilot. No software purchase, blanket pay increase or predicted productivity gain is established by this survey.

A 90-day sequence

Days 0-14: agree scope, owners and capacity; choose one workflow and pilot population; collect operational baselines; review current decision criteria; confidentially verify specific accounts. Record who is eligible so future response rates can be calculated.

Days 15-30: introduce the handover checklist and backup responsibilities; make the relevant criteria and review route accessible; offer feedback conversations; operate the concern/action log.

Days 31-60: review delayed/reworked handovers and decision exceptions; ask whether arrangements are usable; adjust administrative burden and verify whether employees understand the changes.

Days 60-90: repeat selected exact survey items, compare process measures, retain strengths and decide whether to scale, adapt or stop. Report participation and composition at both waves.

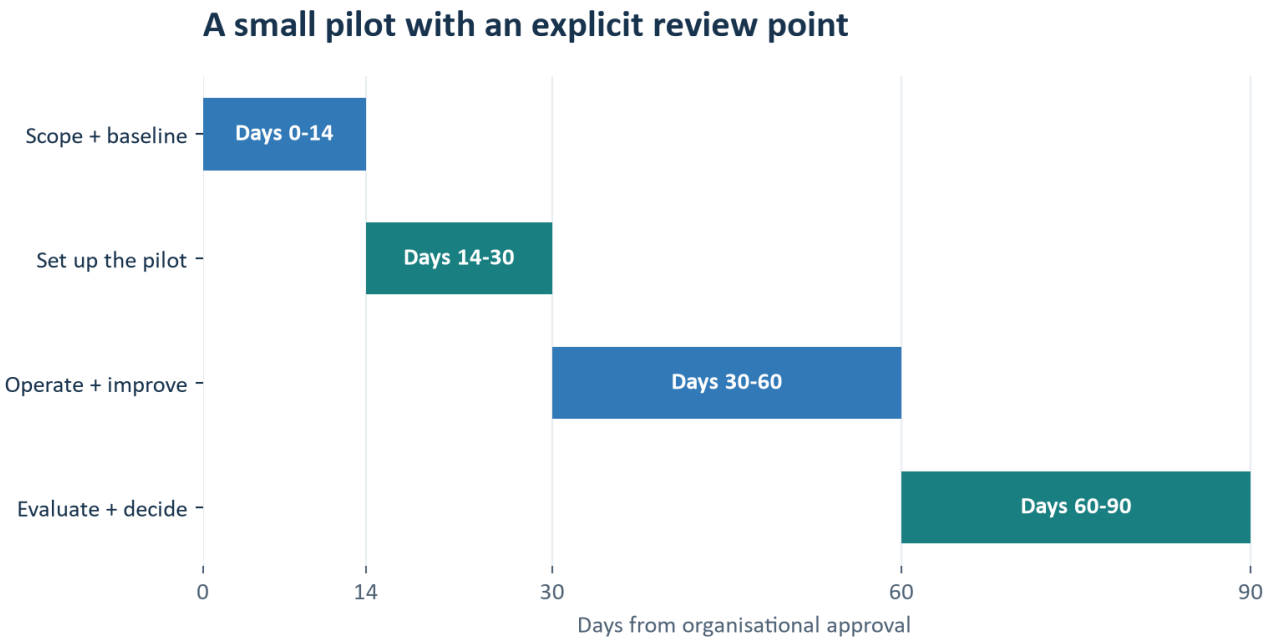


Figure 13. Proposed timing begins at organisational approval. Days 0-14 establish pilot baselines; later phases test operation and employee experience before a scale/adapt/stop decision.

The notebook proposes process standards such as 100% coverage of agreed pilot roles/tasks, at least 95% handover completeness by Day 60, normal-concern acknowledgement within two business days and owner/next-update assignment within five. These are **proposed operating standards**, not observed performance or effect estimates. Five days is not a promise that every concern will be resolved. Management must confirm feasible scope, owners and expectations.

17. How to know whether the company benefited

Implementation and outcomes are different. Publishing criteria is implementation. Employees understanding criteria and experiencing more consistent decisions are outcomes. Completing a feedback conversation is implementation. Employees finding the feedback useful is an outcome.

Use a small monitoring panel: CAM2, FAI3/FAI5, a handover measure, a decision/response measure, and pride/within-team support as guardrails. The larger 19-measure KPI dictionary is a reference, not a requirement to measure every item every week.

For example:

```
Handover completeness = complete in-scope handovers / all logged in-scope handovers × 100
```

```
Criteria access = pilot roles with current accessible criteria and review route / all agreed pilot roles × 100
```

```
Outcome change in percentage points = follow-up favourable % - comparable baseline favourable %
```

An operational KPI needs an agreed eligible denominator. Logging only successful handovers would create a misleading rate. The whole-company response-pool figure is not automatically the baseline for a selected pilot group.

Numerical experience-improvement targets should be agreed after the pilot baseline and feasible sample are known. There is no data-based justification here for promising a particular retention increase, cost saving or return on investment. If the company wants to quantify business benefit, it needs operational volumes, delay/rework definitions, time/cost data and a defensible evaluation design.

A before/after improvement alone does not prove that the pilot caused it. Different respondents, seasonal work, policy changes or other events may contribute. If feasible, stage implementation or retain a relevant comparison group, document selection and compare composition. Even then, assumptions need examination.

18. Reproducibility audit and current limitations

For this explanation, all **73 code cells** were executed against the source workbook in a temporary output directory, keeping the source, notebook and tracked outputs unchanged. The rerun completed after one audit-only portability adjustment described below. It reproduces the headline values and the 327×56 processed schema; all processed rating values and the codebook match the repository versions. All 29 aggregate CSVs were reproduced in content. The network edge CSV differed only in row order among tied support counts and matched after sorting by topic-pair keys.

The notebook's comment hash currently serializes CSV text with the operating-system default line terminator. On Windows that produces CRLF and a different hash from the stored LF-based reference even when comments are identical. Using `to_csv(index=False, lineterminator="\n")` in the temporary audit reproduces the expected hash exactly. The notebook itself still needs that portability adjustment if it is to rerun unmodified on this Windows environment.

Windows text encoding can also matter because several report-writing calls omit an explicit encoding. The audit used Python UTF-8 mode for Unicode report output. A future portability edit can explicitly set UTF-8 on those writes. This is an execution concern, not evidence that the substantive data changed.

The first exploratory chart cells retained some local saved outputs after the earlier pull. This explanation uses recomputed results rather than assuming all cached outputs are current. It does not save a new execution state into the notebook.

Some notebook status messages are older than the later cells: Step 5 mentions the timed roadmap as future work even though Step 6 drafts it. The methodology table contains both an early action-options status and a later completed-draft roadmap row. The consistent current position is: an analytical package and proposed roadmap exist; independent human coding validation, clarification of retained cases, verification of specific accounts, organisational approval, outcome targets and a PowerPoint remain outstanding.

The principal substantive limits are unknown workforce response rate and representativeness, unverified unique identity, no causal design or observed turnover outcomes, approximate role groups, unvalidated composite scores and assistant-interpreted comments. Those limits define what the results mean; they do not erase the practical value of a measured pilot.

19. Sources and related project outputs

The evidence comes from the supplied master workbook and the notebook's explicit transformations and coding ledger. The audit independently recomputed descriptive values and intersections; it did not independently recode all comments.

- Working notebook
- Source workbook
- Existing data-to-action guide
- Concise synthesis
- Interactive topic network
- Project brief and working decisions

Method definitions were checked against the primary SciPy Spearman documentation, Kruskal-Wallis documentation and false discovery control documentation. BY adjustment controls the expected false discovery proportion under its assumptions, including dependent tests. It does not make every retained finding causal or important. Other research links are maintained in the notebook's final methodology reference.

Appendix: every notebook cell explained

Each entry gives the current cell number, type and stable ID. These are reference cards rather than a dense multi-column table.

Cell numbers below are positions in the current 146-cell notebook, not execution counts shown by Jupyter. The stable cell ID is included so a cell remains identifiable if the notebook is later rearranged. Documentation and output-generation cells are included because they govern how the analysis should be interpreted and reproduced.

Cell 1 / MARKDOWN / `c000`

States the organisational purpose, source, assignment context and notebook structure. Establishes that analysis supports actions and that the supplied survey is not an official validated index.

Cell 2 / CODE / `c001`

Imports libraries, constructs paths and sets display/plot defaults. This creates the tools and constants used later; it does not calculate an employee finding.

Cell 3 / MARKDOWN / `c002`

Introduces the workbook inventory. This separates source selection from immediately analysing whichever sheet opens first.

Cell 4 / CODE / `c003`

Lists every sheet and its dimensions. Counts include header/other rows, so sheet row counts must be interpreted before calling them respondents.

Cell 5 / MARKDOWN / `c004`

Explains which sheet is authoritative and how other sheets are used. Documents the unresolved 330-versus-327 discrepancy rather than guessing a removal reason.

Cell 6 / CODE / `c005`

Inspects raw English tenure values as text to expose date conversion. This is a source-format check, not an analysis of employee tenure trends.

Cell 7 / MARKDOWN / `c006`

Explains the corrupted tenure-band issue and why the selected master is used. A date representation here does not measure a survey date or exact service duration.

Cell 8 / CODE / `c007`

Displays the HR mapping/pivot area after dropping entirely empty rows/columns. It reveals the source of the thematic option mapping and a conflicting pivot total.

Cell 9 / MARKDOWN / `c008`

Separates the reusable option mapping from the unreliable pivot count. The later HR counts are recomputed from actual master records.

Cell 10 / MARKDOWN / `c009`

Explains the master's two-row header, with merged dimension labels above question text.

Cell 11 / CODE / `c010`

Reads those two header rows and transposes a sample for inspection. Transpose turns columns into rows so headers are easier to inspect.

Cell 12 / CODE / `c011`

Forward-fills merged block labels and extracts question text. Forward fill repairs header structure; it is not imputing missing employee ratings.

Cell 13 / MARKDOWN / `c012`

Documents the overwritten designation header and the source check used to recover its meaning. Avoids treating a response value as a variable name.

Cell 14 / CODE / `c013`

Defines demographic/comment names and generates 30 short item codes from block order. The modulo operation numbers items 1-5 within the supplied five-item blocks.

Cell 15 / CODE / `c014`

Builds a codebook with field name, block and full question wording. The explicit designation repair is applied here.

Cell 16 / CODE / `c015`

Loads the master after skipping its two header rows and assigns the 43 defined column names. This creates raw, the main source table.

Cell 17 / CODE / `c016`

Creates reusable item, block, comment and demographic lists. These ensure later selections refer to the intended columns.

Cell 18 / MARKDOWN / `c017`

Introduces source quality checks before interpretation.

Cell 19 / CODE / `c019`

Counts valid and missing values, complete rating cells, names and exact duplicate rows excluding names. Completeness and exact-duplicate absence have limited meanings.

Cell 20 / MARKDOWN / `c020`

Interprets the completeness/name checks and explains removal of the dedicated name field. Does not diagnose answering bias from name supply.

Cell 21 / MARKDOWN / `c023`

Introduces inspection of the original response labels.

Cell 22 / CODE / `c024`

Flattens all 30 rating columns and counts original text categories. The unit here is an item answer, so counts total 9,810 rather than 327.

Cell 23 / MARKDOWN / `c025`

Explains spelling/capitalization/language-form variants and their harmonization. Language equivalence remains a substantive assumption to review.

Cell 24 / CODE / `c026`

Applies the response-label mapping and counts unmapped/missing results. Zero unmapped answers is a valid coding check, not evidence of measurement validity.

Cell 25 / MARKDOWN / `c027`

Introduces response-style checks.

Cell 26 / CODE / `c028`

Creates numeric rating table L and flags rows with one unique category across all 30 items. Also counts the identical category used by flagged records.

Cell 27 / MARKDOWN / `c029`

Explains why all identical-category records remain in the primary analysis and why exclusion is only a sensitivity comparison.

Cell 28 / MARKDOWN / `c031`

Introduces category hygiene for demographics and designation text.

Cell 29 / MARKDOWN / `c034`

Warns that job levels are keyword-based analyst approximations rather than an HR-verified hierarchy.

Cell 30 / CODE / `c035`

Counts source departments. This identifies large reporting groups without assuming the distribution represents the workforce.

Cell 31 / CODE / `c036`

Counts source locations. This exposes the large contribution from Mumbai and small-site disclosure issues.

Cell 32 / MARKDOWN / `c037`

Introduces the cleaned analysis table df.

Cell 33 / CODE / `c038`

Defines ordered designation keyword rules, city-to-region mapping and tenure order. Earlier keyword matches take precedence over later ones.

Cell 34 / CODE / `c039`

Drops names, replaces rating text with numbers, creates reporting groups and response-context flags, and orders tenure categories. Source records stay in the same order for later ledger linkage.

Cell 35 / CODE / `c040`

Checks unmatched region values and job-level counts. Any unmapped city would need attention before regional comparisons.

Cell 36 / CODE / `c041`

Lists ambiguous designation categories for review. Unclear is retained rather than forced into a seniority level.

Cell 37 / MARKDOWN / `c042`

Defines favourable shares, block means, the 25-item culture mean and the separate 30-item favourable summary. These definitions prevent changes of denominator or construct later.

Cell 38 / CODE / `c043`

Calculates six respondent block means, CULTURE_MEAN and fav_pct. Equal category spacing and equal weighting are working assumptions.

Cell 39 / MARKDOWN / `c045`

Introduces respondent composition rather than workforce representation.

Cell 40 / CODE / `c046`

Draws composition bars for region, role, tenure, gender, department and location. Location displays only the top 10, so it is not a complete location list.

Cell 41 / MARKDOWN / `c047`

Explains sample composition, missing workforce denominators and the provisional minimum-group rule. Small-group suppression does not establish anonymity.

Cell 42 / MARKDOWN / `c051`

Introduces individual-item distributions as the main rating evidence.

Cell 43 / CODE / `c052`

Calculates each category's percentage per question, aligns all five categories and adds question labels. Missing categories receive zero counts/shares, not missing-answer imputation.

Cell 44 / CODE / `c053`

Plots five-category diverging bars. The central split of Sometimes is for drawing the chart and does not redefine that response.

Cell 45 / CODE / `c054`

Builds exact category counts and item statistics, retaining precision until display. Mean, median, favourable, Sometimes, unfavourable and top-box each describe different features.

Cell 46 / MARKDOWN / `c055`

Explains why high overall answers and lower individual experiences can coexist and why small rank differences may not matter.

Cell 47 / MARKDOWN / `c057`

Introduces block summaries and consistency diagnostics.

Cell 48 / CODE / `c058`

Defines Cronbach alpha and calculates block means, pooled favourable shares and alphas. A block's favourable share averages its item-level favourable shares here.

Cell 49 / MARKDOWN / `c059`

Explains the limits of alpha and the lack of established scale validity or dimensional distinctness.

Cell 50 / CODE / `c060`

Computes alpha across all 30 items. The high result is a consistency/overlap diagnostic, not validation of one overall factor.

Cell 51 / CODE / `c061`

Correlates each item with the total of the other four in its block, then shows the eight lowest diagnostics. The code uses Pearson correlation for this check.

Cell 52 / MARKDOWN / `c062`

Explains why an item-total result invites review rather than automatic deletion.

Cell 53 / CODE / `c063`

Shows respondent block-score boxplots. Read centre and spread, and do not label plotted extreme records as invalid without other evidence.

Cell 54 / MARKDOWN / `c064`

Introduces relationships among rating items.

Cell 55 / CODE / `c065`

Computes the 30-by-30 Spearman correlation matrix and plots a heatmap with supplied-block boundaries. The diagonal is self-correlation.

Cell 56 / CODE / `c066`

Averages within-block and all-pair upper-triangle correlations, excluding diagonal self-correlations and duplicate mirrored pairs.

Cell 57 / MARKDOWN / `c067`

Explains that average associations do not establish the number of factors and that factor analysis remains conditional on a useful measurement question.

Cell 58 / MARKDOWN / `c069`

Separates advocacy, intention to stay and satisfaction. Identifies why the recommendation item is not standard eNPS and why stay intention is not turnover.

Cell 59 / CODE / `c070`

Calculates item-outcome Spearman coefficients excluding all three outcomes from predictors. Displays the top 10 by stay-intention association, not causal intervention priorities.

Cell 60 / MARKDOWN / `c072`

Explains shared questionnaire context and the absence of independent or causal effects in bivariate correlations.

Cell 61 / MARKDOWN / `c074`

Defines the provisional reporting/testing rules and one family of 11 overlapping exploratory comparisons. Describes the BY adjustment rationale.

Cell 62 / CODE / `c075`

Defines seg(): aggregate group n/mean/median/favourable answers, suppress small-group scores, and compare eligible distributions with Kruskal-Wallis.

Cell 63 / CODE / `c076`

Displays tenure-group summaries. The p-value returned by seg is not shown here; the joint adjusted table is calculated later.

Cell 64 / CODE / `c077`

Displays approximate role-group summaries, retaining Unclear and suppressing scores for the smallest role group.

Cell 65 / CODE / `c078`

Displays regional summaries and suppresses Overseas scores because its group is below the provisional threshold.

Cell 66 / CODE / `c079`

Displays department-reporting-group summaries. Other is a mixture of smaller departments and must not be treated as one coherent function.

Cell 67 / CODE / `c080`

Displays gender summaries and suppresses the very small non-disclosure group. Descriptive differences do not establish discrimination or a population effect.

Cell 68 / MARKDOWN / `c082`

Explains the optional-name comparison as a response-context check and lists possible alternative explanations.

Cell 69 / CODE / `c083`

Runs six full-sample and five non-straightliner comparisons, then adjusts all 11 p-values jointly with BY. Eligibility can change after removing records.

Cell 70 / CODE / `c084`

Draws working block means by tenure with group sizes and suppression. It is a descriptive visualization, not a longitudinal tenure effect.

Cell 71 / CODE / `c085`

Draws working block means by department with group sizes and suppression. Compare with the adjusted tests before claiming group differences.

Cell 72 / MARKDOWN / `c086`

Introduces the exclusion sensitivity and explains that removing records changes the sample rather than correcting toward a known truth.

Cell 73 / CODE / `c087`

Shows block means with/without straight-liners and the signed difference. A negative difference indicates lower means in the alternative sample.

Cell 74 / MARKDOWN / `c091`

Introduces the single selected HR priority, separate from written-answer themes.

Cell 75 / CODE / `c092`

Loads the 10-option-to-eight-theme mapping, strips theme whitespace and verifies that every HR option maps successfully.

Cell 76 / CODE / `c093`

Plots all 10 original HR options by response count. This preserves the distinction between listening and keeping employees informed.

Cell 77 / CODE / `c094`

Shows shares for eight mapped HR themes. Rounding occurs before multiplication here, so the displayed theme percentages have approximately one-decimal precision.

Cell 78 / CODE / `c095`

Constructs demographic-by-HR contingency tables, computes descriptive Cramer's V, audits expected cells and withholds routine p-values when any expected count is below five.

Cell 79 / MARKDOWN / `c098`

Explains single-choice limits and why sparse tables do not justify a simple departmental significance claim.

Cell 80 / MARKDOWN / `c099`

Introduces comment coverage and the need to interpret brief answers in their question context.

Cell 81 / CODE / `c100`

Creates stripped string comment table txt and profiles nonblank counts, blanks, review-pattern matches and median answer length. The placeholder regex flags review; it does not delete comments.

Cell 82 / MARKDOWN / `c104`

Explains why repeated or polished wording does not prove dishonesty, automation or lack of meaning.

Cell 83 / CODE / `c105`

Counts records participating in repeated-text groups per question. This is a wording flag; distinct people can legitimately give identical brief replies.

Cell 84 / MARKDOWN / `c106`

Introduces processed exports.

Cell 85 / CODE / `c107`

Saves the cleaned survey and codebook. Running this cell normally writes tracked processed files; the audit redirected output to temporary folders.

Cell 86 / MARKDOWN / `c108`

Summarizes the initial EDA with caveats. Later steps extend these findings and supersede some early descriptions of work as pending.

Cell 87 / MARKDOWN / `step1-heading`

Introduces the decision-first analytical foundation and identifies missing organisational context and causal limitations.

Cell 88 / MARKDOWN / `step1-question-map`

Maps business questions to exact survey evidence, suitable analyses and possible decisions. This prevents method selection from becoming detached from the company's purpose.

Cell 89 / MARKDOWN / `step1-base-heading`

Defines response records versus item-answer units and explains the foundational audit.

Cell 90 / CODE / `step1-base-audit`

Asserts unique item codes, consistent row order, valid ratings, removed names, reconciled categories and complete HR mapping; then displays the base register.

Cell 91 / MARKDOWN / `step1-measures`

Records the measurement contract: definitions, units, denominators and interpretation limits. This governs consistent comparisons across later outputs.

Cell 92 / MARKDOWN / `step1-research`

Explains the methods' research basis and distinguishes practitioner discussion from primary methodological support. Source relevance does not itself validate this dataset.

Cell 93 / MARKDOWN / `step1-fuzzy`

Explains graded membership and distinguishes fuzzy approaches before checking a specific frequency-based construction.

Cell 94 / CODE / `step1-fuzzy-check`

Reconstructs block scores from category frequencies and equal weights, then asserts equality with ordinary means. It demonstrates no additional information in this specific construction.

Cell 95 / MARKDOWN / `step1-fuzzy-decision`

Defers a calibrated fuzzy model until a real decision and justified memberships/weights exist. Does not claim all fuzzy methods are equivalent to a mean.

Cell 96 / MARKDOWN / `step1-hr-concepts`

Defines useful interpretive lenses such as justice, employee voice and psychological safety, while avoiding claims that those full constructs were validated by this questionnaire.

Cell 97 / MARKDOWN / `step1-next`

Connects the foundation to the later descriptive and comment analyses. Some future-tense text reflects the sequence in which the notebook was built.

Cell 98 / MARKDOWN / `step2-heading`

Compares candidate methods by the decision they answer. Chooses contextual coding, phrases, paired profiles and integration while leaving unnecessary advanced models conditional.

Cell 99 / MARKDOWN / `step2-item-heading`

Defines favourable-share ranking and explains why not favourable includes Sometimes rather than automatically meaning dissatisfied.

Cell 100 / CODE / `step2-item-profile`

Adds exact lower-category counts and non-straightliner shares, exports the item evidence table and plots six lower favourable shares beside six strengths.

Cell 101 / MARKDOWN / `step3-heading`

Describes the contextual comment-reading/coding workflow, exploratory status, review coverage and remaining independent validation needs.

Cell 102 / CODE / `step3-codebook`

Defines all 15 topics with inclusion/exclusion boundaries and paraphrased examples. This gives each letter in the ledger an explicit meaning.

Cell 103 / CODE / `step3-ledger`

Stores the original semantic assignments for every nonblank answer by question and source-row position. It is fixed interpretive input, not a trained classifier or a verified employee identity list.

Cell 104 / MARKDOWN / `step3-review-note`

Explains the same-assistant re-review cohorts and why this is separate from independent human validation.

Cell 105 / CODE / `step3-assistant-rereview`

Preserves first-pass assignments, identifies review cohorts, applies explicit changes with reasons and reconstructs the current ledger. Cohorts overlap; their counts are not additive.

Cell 106 / CODE / `step3-parse-and-audit`

Checks the text/order hash, parses markers/topics, verifies every nonblank record is represented once, resolves documented references, adds review flags and produces coverage totals. Its default-line-ending hash needs the Windows portability fix described above.

Cell 107 / MARKDOWN / `step3-count-contract`

Defines unique-record request counts and preserve-prompt strength counts. Prevents adding overlapping topics or subtracting counts from different prompt contexts.

Cell 108 / CODE / `step3-topic-counts`

Builds requester/preserve counts, alternative denominators, flagged-case sensitivity, per-question topics and wording diversity. Exports a topic figure while retaining all exact repeated replies in main counts.

Cell 109 / CODE / `step3-context-and-fairness`

Summarizes fairness stances, including no-concern replies, and plots the question-specific topic heatmap. This makes prompt dependence visible.

Cell 110 / MARKDOWN / `step3-ngram-heading`

Defines phrase boundaries, unique-record support and privacy filters. The numeric-token caveat in the walkthrough qualifies literal adjacency claims.

Cell 111 / CODE / `step3-ngram-analysis`

Tokenizes eligible answers, excludes prompt/boilerplate/name-related phrases, counts supported exact sequences and displays top phrases per question/width. It supports retrieval rather than semantic coding.

Cell 112 / MARKDOWN / `step4-heading`

Introduces integration using the same response records rather than treating separate percentages as unrelated populations.

Cell 113 / CODE / `step4-paired-and-bridge`

Cross-tabulates within/cross-team favourability and joins topic-requester sets with related item answers. The bridge shows why favourable ratings and improvement requests can coexist.

Cell 114 / MARKDOWN / `step4-bridge-limit`

Explains the broad topic-to-specific item mismatch and why the bridge neither diagnoses hidden dissatisfaction nor establishes causes.

Cell 115 / CODE / `step4-tenure-profile`

Follows up tenure with culture, growth, stay intention, recognition and selected topic requests; applies count/group suppression and retains response-style sensitivity.

Cell 116 / MARKDOWN / `step4-prioritisation`

Separates reach, seriousness, evidence and feasibility. Gives specific unverified accounts a confidential verification route rather than letting frequency alone determine urgency.

Cell 117 / CODE / `step4-evidence-register`

Creates seven evidence-to-action rows linking baseline, comments, proposed action, owner, measure and limitation. These are options for review rather than approved organisational commitments.

Cell 118 / CODE / `step4-story-summary`

Displays a narrative about strengths, consistent practices, cross-team execution, follow-through and verification. It summarizes earlier evidence rather than adding a new calculation.

Cell 119 / CODE / `step4-export`

Exports 11 aggregate reference tables and checks ledger coverage, unique question/record keys and valid topics. The displayed output-count message reflects the outputs at that stage.

Cell 120 / MARKDOWN / `step5-heading`

Records the selected cross-team/fairness focus and three supporting themes. Makes explicit that selection was a project decision rather than a synthetic statistical importance score.

Cell 121 / MARKDOWN / `step5-cross-heading`

Distinguishes within-team support, collaboration across teams and access to other teams before examining paired results.

Cell 122 / CODE / `step5-cross-depth`

Calculates complete CAM item profiles, favourable discordance, original-category comparisons and teamwork-prompt topic counts. Provides precise baselines for the cross-team interpretation.

Cell 123 / MARKDOWN / `step5-cross-reading`

Explains 45-versus-16 discordance and the relevant comment context. Warns that removing straight-liners cannot independently validate a group they could never enter.

Cell 124 / MARKDOWN / `step5-fair-heading`

Separates five dimensions of practical fairness and explains why any 1-3 pattern is not an unfair-treatment diagnosis.

Cell 125 / CODE / `step5-fair-depth`

Calculates fairness item categories, breadth, explicit negative answers, no. of overlapping cross-team/fairness records and selected group profiles. Applies focused n/count suppression.

Cell 126 / MARKDOWN / `step5-fair-reading`

Interprets the fairness counts and retains no-concern experiences. Explains why overlapping cross-team and fairness issues need different follow-up and do not identify a culprit department.

Cell 127 / MARKDOWN / `step5-supporting`

Links recognition, roles/workload/backup and voice/follow-through to the two priorities while preserving pride and treating specific accounts separately.

Cell 128 / MARKDOWN / `step5-network-method`

Defines same-answer nodes/edges, association normalization, display thresholds, custom layout/clustering and prompt/parameter limits.

Cell 129 / CODE / `step5-network-functions`

Defines pair counting, normalized affinity, greedy modularity and seeded force layout, then embeds the local interactive HTML. Most long text in the cell is interface code rather than new analysis.

Cell 130 / CODE / `step5-network-build`

Builds the primary/excluded-question networks, coordinates, labels and parameter checks; exports HTML/JSON, static PNG/SVG and VOSviewer-compatible files. Primary coordinates/colours stay fixed across exclusion views.

Cell 131 / MARKDOWN / `step5-network-reading`

Explains key pairs and the large share driven by individual prompts. Prevents the graph being presented as independent or causal evidence.

Cell 132 / CODE / `step5-final-package`

Consolidates five workstreams, exports focused/network tables, writes the synthesis report and asserts headline counts and finite layout values. Some evidence strings are hard-coded and should be refreshed if data/coding change.

Cell 133 / MARKDOWN / `step6-heading`

Introduces the review audit, implementation pathway and generated guide. These later cells update earlier pending-work messages.

Cell 134 / CODE / `step6-review-audit`

Summarizes rereads and before/after counts, recomputes 14 expected facts, exports review/verification tables and separates traceability from coding accuracy.

Cell 135 / MARKDOWN / `step6-review-reading`

Explains what the 568 rereads and 53 changes mean, including remaining 75 clarification cases and absence of independent human validation.

Cell 136 / MARKDOWN / `step6-roadmap-intro`

Defines a small pilot starting at organisational approval and distinguishes the whole response pool from the selected pilot baseline.

Cell 137 / CODE / `step6-roadmap`

Creates 13 proposed actions with timing, accountable roles, deliverables, standards and dependencies, then exports the roadmap and a timeline graphic. It records a proposal, not completion.

Cell 138 / CODE / `step6-kpis`

Creates 19 outcome/process/guardrail definitions with denominators, baseline needs, owner roles and target-setting rules; adds Day-0/14/30/60/90 decisions and exports both references.

Cell 139 / MARKDOWN / `step6-measurement`

Explains implementation versus experience outcomes, exact repeated wording, response composition and limits of a simple before/after comparison.

Cell 140 / CODE / `step6-complete-guide`

Generates the comprehensive data-to-action Markdown report using current objects and tables, and updates the synthesis reference. Its long size largely reflects report-writing prose rather than another statistical model.

Cell 141 / MARKDOWN / `c109`

Marks the annexure, separating the analytical sequence from reference material.

Cell 142 / MARKDOWN / `c110`

Defines abbreviations and technical terms used in the notebook, including p-values, reliability, percentage points and the meaning of the working summaries.

Cell 143 / MARKDOWN / `c112`

Documents all created fields in the 56-column processed dataset and notes why removal of the name field does not make comments anonymous.

Cell 144 / MARKDOWN / `c113`

Lists exact response, designation, region and HR mappings. It is the reference for reproducing transformations and judging approximations.

Cell 145 / MARKDOWN / `c114`

Records source lineage, the unresolved source discrepancy, transformations, outputs and environment expectations. The raw workbook remains the preserved input.

Cell 146 / MARKDOWN / `methodology-reference`

Maintains the final method inventory with purpose, inputs, assumptions, current status and limits. Distinguishes executed analyses, proposed actions and conditional methods; read later roadmap status alongside older action-options wording.